ESTIMATING THE NUMBER OF ZEROS OF DEDEKIND ZETA-FUNCTIONS

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Abstract

Abstract. In this article, I derive a new approach to estimate the number of non-trivial zeros of a given Dedekind zeta function with absolute height at most $T \geq 1$ counted with multiplicity. The error term in corresponding asymptotic formula improves all previous results, even in the case of the Riemann zeta function.

1 Introduction

In this article, we consider Dedekind zeta functions associated with a given algebraic number field. Let K be an algebraic number field of degree n_K then the associated Dedekind zeta-function is given by

$$\zeta_{\mathrm{K}}(s) = \sum_{I \subset \mathcal{O}_K \setminus 0} N_K(I)^{-s}, \tag{1.1}$$

for $\mathrm{Re}(s) > 1$ [Neu99, Definition 5.1], where the sum is taken over all non-zero integer ideals of K. Similarly to how the Riemann zeta function is related to the distribution of primes, the Dedekind zeta functions are closely related to the distribution of prime ideals. Furthermore, the Dedekind zeta function has properties very similar to the Riemann zeta function, in that the definition in (1.1) allows an analytic continuation to the entire complex plane which has a simple pole at 1 and satisfies a functional equation. We have

$$\xi_K(s) := s(s-1) d_K^{s/2} \gamma_K(s)\, \zeta_{\mathrm{K}}(s), \tag{1.2}$$

where

$$\gamma_K(s) = \left(\pi^{-\frac{s+1}{2}} \Gamma\left(\frac{s+1}{2}\right)\right)^{r_2} \left(\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right)\right)^{r_1+r_2}. \tag{1.3}$$

Here, d_K denotes the absolute discriminant of the number field K. Furthermore, r_1 and r_2 denote the number of real and complex places of $K/\mathbb{Q}$. So, in particular, we have $n_K = r_1 + 2r_2$. Due to computational advantages, we will also use this representation for the gamma factor

$$\gamma_K(s) = \gamma_1(s)^{r_1} \gamma_2(s)^{r_2} \tag{1.4}$$

with

$$\gamma_1(s) := \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \quad \text{and} \quad \gamma_2(s) := \pi^{-s} 2^{1-s} \Gamma(s),$$

which we may obtain from equation (1.3) by Legendre's duplication formula [WW20, Corollary 12.1.3] for the Gamma function. The completed Dedekind zeta function satisfies the functional equation [Neu99, Corollary 5.10] as displayed below,

$$\xi_K(s) = \xi_K(1-s). \tag{1.5}$$

In this context, we strive to study the distribution of the zeros of $\zeta_{\mathrm{K}}(s)$. From the Euler product for the Dedekind zeta function $\zeta_{\mathrm{K}}(s)$ [Neu99, Proposition 5.2]

$$\zeta_{\mathrm{K}}(s) = \prod_{\mathfrak{p}\subset\mathcal{O}_K} \left(1 - N(\mathfrak{p})^{-s}\right)^{-1} \tag{1.6}$$

we can see that $\zeta_{\mathrm{K}}(s)$ and, by extension $\xi_K(s)$ have no zeros with $\mathrm{Re}(s) > 1$ and by the functional equation we can tell that all zeros of $\xi_K(s)$ must be located inside the critical strip with $\mathrm{Re}(s) \in [0,1]$. We call these the non-trivial zeros of $\zeta_{\mathrm{K}}(s)$ and throughout this paper we will denote them by $\rho = \beta + i\omega$, where β denotes the real part of the zero and ω denotes the imaginary component. In this article, we investigate the horizontal distribution of the non-trivial zeros. Our goal is to estimate the quantity $N_K(T)$ given by

$$\begin{aligned} N_K(T) :=& \#\left\{\rho \in \mathbb{C} \mid \zeta_{\mathrm{K}}(\rho) = 0,\ 0 \le \beta \le 1,\ |\omega| \le T\right\}' \\ =& \frac{1}{2}\#\left\{\rho \in \mathbb{C} \mid \zeta_{\mathrm{K}}(\rho) = 0,\ 0 \le \beta \le 1,\ |\omega| \le T\right\} \\ &+ \frac{1}{2}\#\left\{\rho \in \mathbb{C} \mid \zeta_{\mathrm{K}}(\rho) = 0,\ 0 \le \beta \le 1,\ |\omega| < T\right\} \end{aligned}$$

counted with multiplicity. Here, the prime notation indicates that ζ_{K}-zeros with imaginary part T are counted with half their weight. Following and improving on the work of Backlund [Bac16], McCurley [McC84], Rosser [Ros41], Kadiri and Ng [KN12], Hasanalizade, Shen and Wong [HSW21] were able to prove that

$$\left|N_K(t) - \frac{T}{\pi}\log\left(d_K\left(\frac{T}{2\pi e}\right)^{n_K}\right)\right| \le C_1\left(\log d_K + n_K \log T\right) + C_2 n_K + C_3$$

for $T \ge T_0$ and admissible tuples (C_1, C_2, C_3, T_0). The table below contains such admissible tuples.

	$T \ge 1$		$T \ge 10$	
C_1	C_2	C_3	C_2	C_3
0.22737	23.02528	4.51954	22.97204	3.30668
0.24493	6.66558	4.21201	6.60397	3.12362
0.26304	5.22032	4.08149	5.15251	3.05074
0.28032	4.43521	4.00936	4.36214	3.01124
0.29590	3.93889	3.96852	3.86136	2.98903

Table 1: Admissible choices for (C_1, C_2, C_3), taken from [HSW21]

Similarly, more refined estimates exist for the Riemann Zeta-function. In this case, we define the zero counting function $N(T)$ as the number of zeros with real part between 0 and 1 and imaginary part between 0 and T. We then have that

$$\left|N(T) - \frac{T}{2\pi}\log\left(\frac{T}{2\pi e}\right)\right| \le C_1 \log(T) + C_2 \log\log(T) + C_3$$

for $T \ge T_0$. The table below summarizes the advances made for this estimate.

	C_1	C_2	C_3	T_0
von Mangoldt [Man05] (1905)	0.4320	1.9167	13.0788	28.5580
Grossmann [Gro13] (1913)	0.2907	1.7862	7.0120	50
Backlund [Bac16] (1918)	0.1370	0.4430	5.2250	200
Rosser [Ros41] (1941)	0.1370	0.4430	2.4630	2
Trudgian [Tru12] (2012)	0.1700	0	2.8730	e
Trudgian [Tru14] (2014)	0.1120	0.2780	3.3850	e
Platt–Trudgian [PT15] (2015)	0.1100	0.2900	3.165	e
Hasanalizade, Shen, and Wong [HSW22] (2022)	0.1038	0.2573	9.4925	e
Bellotti–Wong [BW25](2025)	0.10076	0.24460	8.08344	e

Table 2: Table with advances made for this estimate, taken from [BW25]

2 Results

Inspired by Turing's method [Tur53, Lemma 1], we take a different approach to estimate this quantity, yielding the following result.

Theorem 2.1. *Let K be a number field with degree n_K and absolute discriminant d_K. Then*

$$\left|N_K(T) - \frac{T}{\pi}\log\left(d_K\left(\frac{T}{2\pi e}\right)^{n_K}\right)\right| \leq C_1\left(\log d_K + n_K\log T\right) + C_2 n_K + C_3$$

for $T \geq 1$, where $(C_1, C_2, C_3) = (0.194, 8.161, 2.001)$.

Further admissible tuples (C_1, C_2, C_3) are given in table (4). As a corollary of Theorem 2.1 we obtain the following result.

Corollary 2.2. *Let $N(T)$ be defined by $\#\{\rho \in \mathbb{C} \mid \zeta(\rho) = 0,\ \mathrm{Re}(\rho) \in [0,1],\ \mathrm{Im}(\rho) \in [0,T]\}'$ counted with multiplicity and such that zeros with imaginary part T are counted with half weight. Then*

$$\left|N(T) - \frac{T}{2\pi}\log\left(\frac{T}{2\pi e}\right)\right| \leq C_1 \log T + C_2$$

for $T \geq 1$, where $(C_1, C_2) = (0.097, 5.081)$.

Further admissible tuples are given in table (6). These results improve on previous results in all ranges of T, when $T \geq 5$. The calculations for this paper were done with python and the code can be found at `https://arxiv.org/abs/2510.27444`. The code takes a set of parameters to be defined in Lemma 3.1, tests whether these are admissible, and then computes the related constants (C_1, C_2, C_3) for Theorem 2.1. With this method, different tuples (C_1, C_2, C_3) can be calculated to optimize the estimate around specific ranges for T. The values given in Theorem 2.1 and Corollary 2.2 are optimized for asymptotic behavior when $T \to \infty$.

3 Proof of Theorem 1

We consider a Dedekind zeta function $\zeta_{\mathrm{K}}(s)$ and want to estimate $N_K(T)$. Let us first only consider $T \in \mathbb{R}$ such that T is not the imaginary part of a ζ_{K}-zero. In the case where T is the ordinate of a ζ_{K}-zero, set

$$N_K(T) = \lim_{\varepsilon \to 0} \frac{N_K(T+\varepsilon) + N_K(T-\varepsilon)}{2}. \tag{3.1}$$

Now, notice that the non-trivial zeros of ζ_{K} are precisely the zeros of $\xi_K(s) = s(s-1)d_K^{s/2}\gamma_K(s)\,\zeta_{\mathrm{K}}(s)$ and all zeros lie inside the critical strip with real part between 0 and 1. This function is holomorphic and thus we can apply the argument principle [FL94, Lemma 7.1] to compute the number of zeros. We have

$$N_K(T) = \frac{1}{2\pi i} \oint_C \frac{\xi_K'(s)}{\xi_K(s)} ds, \tag{3.2}$$

where C is any contour that surrounds the rectangle $0 \pm iT$, $1 \pm iT$ in the counter clockwise direction. Let $d > \frac{1}{2}$, which will be fixed later and define C as the counter that connects the four points $\frac{1}{2} \pm d \pm iT$. Notice that by the functional equation (1.5) and the symmetry of the complex conjugate (3.2) simplifies to

$$N_K(T) = \frac{2}{\pi i} \oint_{C'} \frac{\xi_K'}{\xi_K}(s) ds = \frac{2}{\pi i} \int_{\frac{1}{2}+d}^{\frac{1}{2}+d+iT} \frac{\xi_K'}{\xi_K}(s) ds + \frac{2}{\pi i} \int_{\frac{1}{2}+d+iT}^{\frac{1}{2}+iT} \frac{\xi_K'}{\xi_K}(s) ds,$$

where C' is the path that starts at $s = \frac{1}{2} + d$, goes to $\frac{1}{2} + d + iT$ and then to $\frac{1}{2} + iT$. Because $\frac{1}{2} + d > 1$, we may compute the first integral with (1.2) and the Dirichlet series for $\frac{\zeta_{\mathrm{K}}'}{\zeta_{\mathrm{K}}}(s)$. To this end we want to expand the integral by a copy where the real part is shifted by d such that the entire integral is contained in the right half plane of absolute convergence. This shift enables us to apply an absolutely convergent sum involving the non-trivial ζ_{K}-zeros to evaluate the integral. This approach is inspired by Turing's method [Tur53, Lemma 1]. We obtain

$$\begin{aligned} N_K(T) =& \frac{2}{\pi} \int_{\frac{1}{2}+d+iT}^{\frac{1}{2}+iT} \operatorname{Im}\left(\frac{\xi_K'}{\xi_K}(s)\right) ds + \frac{2}{\pi} \operatorname{Im}\log\left(\xi_K\left(\frac{1}{2} + d + iT\right)\right) \\ =& -\frac{2}{\pi} \int_{\frac{1}{2}}^{\frac{1}{2}+d} \operatorname{Im}\left(\frac{\xi_K'}{\xi_K}(\sigma + iT)\right) - \operatorname{Im}\left(\frac{\xi_K'}{\xi_K}(\sigma + d + iT)\right) d\sigma \\ & - \frac{2}{\pi} \int_{\frac{1}{2}+d}^{\frac{1}{2}+2d} \operatorname{Im}\left(\frac{\xi_K'}{\xi_K}(\sigma + iT)\right) d\sigma + \frac{2}{\pi} \operatorname{Im}\log\left(\xi_K\left(\frac{1}{2} + d + iT\right)\right) \\ =& -\frac{2}{\pi} \int_{\frac{1}{2}}^{\frac{1}{2}+d} \operatorname{Im}\left(\frac{\xi_K'}{\xi_K}(\sigma + iT)\right) - \operatorname{Im}\left(\frac{\xi_K'}{\xi_K}(\sigma + d + iT)\right) d\sigma + E_1(\xi_K)(T), \end{aligned} \tag{3.3}$$

where $E_1(f)$ is an operator given by

$$E_1(f)(t) = \frac{2}{\pi}\left(2 \operatorname{Im}\log\left(f\left(\frac{1}{2} + d + it\right)\right) - \operatorname{Im}\log\left(f\left(\frac{1}{2} + 2d + it\right)\right)\right),$$

for a fixed constant $d > \frac{1}{2}$. Here, $\operatorname{Im}\log(f)$ is interpreted as the continuous variation of the argument of the function f starting from the real axis. We use the notation $\operatorname{Im}\log(f)$ without brackets around the logarithm to signal this difference. Next, we need the following representation for the Dedekind zeta-function to express the remaining integral in (3.3) via a sum involving the non-trivial ζ_{K}-zeros. Because

ξ_K is an entire function of order 1, we can apply Theorem 5.52 from [IK04, page 150] and obtain that

$$\xi_K(s) = e^{Q(s)} \prod_\rho \left(1 - \frac{s}{\rho}\right) e^{\frac{s}{\rho}},$$

where $Q(s)$ is a polynomial of degree at most 1. Hence,

$$\frac{\xi_K'}{\xi_K}(s) = Q'(s) + \sum_\rho \frac{1}{\rho} + \frac{1}{s-\rho} \tag{3.4}$$

where the last sum is absolutely convergent, because there are $O_K(T\log(T))$ terms in the sum in equation (3.4) and

$$\left|\frac{1}{\rho} + \frac{1}{s-\rho}\right| = \left|\frac{s}{\rho(s-\rho)}\right| = \frac{|s|}{|\rho||s-\rho|} = O_s\left(|\rho|^{-2}\right) = O_s(\omega^{-2})$$

for large $\omega = \mathrm{Im}(\rho)$. Notice that because $Q(s)$ is a first degree polynomial, $Q'(s)$ is constant. Furthermore, $\frac{\xi_K'}{\xi_K}(s)$ commutes with complex conjugation, as does the sum in equation (3.4) because the ζ_K-zeros ρ occur in complex conjugate pairs. Therefore, $Q'(s) = Q'(\bar{s})$ and $\mathrm{Im}(Q'(s)) = 0$. Furthermore, we can compute the real part of $Q'(s)$ by exploiting the symmetry of the ξ_K-zeros as well as the functional equation (1.5). Because $Q'(s)$ is constant equations (3.4) and the functional equation imply

$$Q'(s) = \frac{\xi_K'}{\xi_K}(0) = -\frac{\xi_K'}{\xi_K}(1) = -Q'(s) - \sum_\rho \frac{1}{\rho} + \frac{1}{1-\rho} = -Q'(s) - 2\sum_\rho \mathrm{Re}\left(\frac{1}{\rho}\right). \tag{3.5}$$

From this we obtain that $Q'(s) = \sum_\rho \mathrm{Re}\left(\frac{1}{\rho}\right)$. We conclude that

$$\mathrm{Im}\left(\frac{\xi_K'}{\xi_K}(s)\right) = \sum_\rho \frac{\omega - t}{(\sigma-\beta)^2 + (t-\omega)^2} - \frac{\omega}{\beta^2+\omega^2},$$

where we again use the notation that $\rho = \beta + i\omega$. With this tool at our disposal, we will find that we can bound the integral in (3.3) using operators similar to E_1 applied to ξ_K and evaluated at points with real parts greater than 1. Eventually, we will evaluate all of these operators together to take advantage of potential cancellations between these terms. Thus, we obtain

$$\begin{aligned} N_K(T) =& \frac{2}{\pi}\int_{\frac{1}{2}}^{\frac{1}{2}+d} \sum_\rho \frac{T-\omega}{(\sigma-\beta)^2 + (T-\omega)^2} - \frac{T-\omega}{(\sigma+d-\beta)^2+(T-\omega)^2} d\sigma \\ &+ E_1(\xi_K)(T,d) \\ =& \frac{2}{\pi}\sum_\rho g(T,\rho) + E_1(\xi_K)(T), \end{aligned} \tag{3.6}$$

where

$$\begin{aligned} g(t,\rho) &= \int_{\frac{1}{2}}^{\frac{1}{2}+d} \frac{t-\omega}{(\sigma-\beta)^2+(t-\omega)^2} - \frac{t-\omega}{(\sigma+d-\beta)^2+(t-\omega)^2} d\sigma \\ &= -\arctan\left(\frac{\frac{1}{2}+2d-\beta}{t-\omega}\right) + 2\arctan\left(\frac{\frac{1}{2}+d-\beta}{t-\omega}\right) - \arctan\left(\frac{\frac{1}{2}-\beta}{t-\omega}\right), \end{aligned} \tag{3.7}$$

for a given fixed constant $d > \frac{1}{2}$ and again we used the notation $\rho = \beta + i\omega$. Here we use the fact that we consider the finite integral of an absolutely convergent sum to swap the order of summation and integration. Now, due to the functional equation for $\zeta_{\mathrm{K}}(s)$ and the fact that $\zeta_{\mathrm{K}}(s)$ commutes with complex conjugation, we obtain $\zeta_{\mathrm{K}}(1-\bar{\rho}) = 0$ for every ζ_{K}-zero ρ. Furthermore, we observe that the sum in equation (3.6) is again absolutely convergent. Therefore, we may reorder terms and thus group terms associated with ρ and $1-\bar{\rho}$ together. Doing so allows us to cancel the last term from equation (3.7), which yields

$$N_K(T) = \frac{1}{\pi}\sum_{\rho} f(b, T-\omega) + E_1(\xi_K)(T), \tag{3.8}$$

where

$$\begin{aligned} f(b,t) =& 2\arctan\left(\frac{b+d}{t}\right) + 2\arctan\left(\frac{-b+d}{t}\right) \\ & - \arctan\left(\frac{b+2d}{t}\right) - \arctan\left(\frac{-b+2d}{t}\right), \end{aligned}$$

for a given fixed constant $d > \frac{1}{2}$. Here $b := \frac{1}{2} - \beta$. Notice that this defines a harmonic function in b and t for $t \neq 0$. Now, the idea is to limit the contribution of each term in equation (3.8) by a linear combination of poles of different degrees located at $\frac{1}{2} + d + iT - \rho$. We may then evaluate the sum of these poles due to their connection to the logarithmic derivatives of $\xi_K(\frac{1}{2} + d + iT)$. This idea was inspired by a very similar method used by Turing to estimate $S_1(T)$ [Tur53]. The following lemma presents us with the required bounds for $f(b,t)$.

Lemma 3.1. *Let $|b| \leq \frac{1}{2}$ and $t \neq 0$. Then*

$$\begin{aligned} f(b,t) \leq& \frac{\pi}{4}\Bigg[da_1\left(\frac{d+b}{(d+b)^2+t^2} + \frac{d-b}{(d-b)^2+t^2}\right) \\ & + d^2a_2\left(\frac{(d+b)^2-t^2}{((d+b)^2+t^2)^2} + \frac{(d-b)^2-t^2}{((d-b)^2+t^2)^2}\right) \\ & + a_3\left(\frac{2(d+b)t}{((d+b)^2+t^2)^2} + \frac{2(d-b)t}{((d-b)^2+t^2)^2}\right)\Bigg] \quad \text{and} \\ f(b,t) \geq& -\frac{\pi}{4}\Bigg[da_1\left(\frac{d+b}{(d+b)^2+t^2} + \frac{d-b}{(d-b)^2+t^2}\right) \\ & + d^2a_2\left(\frac{(d+b)^2-t^2}{((d+b)^2+t^2)^2} + \frac{(d-b)^2-t^2}{((d-b)^2+t^2)^2}\right) \\ & - a_3\left(\frac{2(d+b)t}{((d+b)^2+t^2)^2} + \frac{2(d-b)t}{((d-b)^2+t^2)^2}\right)\Bigg] \end{aligned}$$

for $d = 0.722$, $a_1 = 1.07$, $a_2 = 0.93$ and $a_3 = 0.365$.

Proof. We prove the upper bound first and conclude the lower bound by symmetry. Let

$$\begin{aligned}H(b,t) := f(b,t) - \frac{\pi}{4}\Bigg[& da_1\left(\frac{d+b}{(d+b)^2+t^2}+\frac{(d-b)}{(d-b)^2+t^2}\right)\\ &+d^2a_2\left(\frac{(d+b)^2-t^2}{((d+b)^2+t^2)^2}+\frac{(d-b)^2-t^2}{((d-b)^2+t^2)^2}\right)\\ &+a_3\left(\frac{2(d+b)t}{((d+b)^2+t^2)^2}+\frac{2(d-b)t}{((d-b)^2+t^2)^2}\right)\Bigg].\end{aligned}$$

Notice that the functions in the brackets may be expressed as the real and imaginary parts of $\frac{1}{d+b+it}$ and $\frac{1}{(d+b+it)^2}$, therefore, they are harmonic functions in b and t for $d+b+it \neq 0$ and a fixed d. For fixed a_1, a_2, and a_3, $H(b,t)$ is then a harmonic function for $t \neq 0$ because $f(b,t)$ is a harmonic function for $t \neq 0$. Notice that it suffices to show that

$$H(b,t) \leq 0 \tag{3.9}$$

for all $(b,t) \in \left[-\frac{1}{2},\frac{1}{2}\right] \times (\mathbb{R} \setminus 0)$ in order to prove the lemma. First, notice that we obtain from the Taylor series of $\arctan(x)$ that

$$H(b,t) = \frac{d^2\pi}{2t^2}(a_2-a_1) + O_b(t^{-3}). \tag{3.10}$$

This verifies the inequality (3.9) for large $|t|$, say $|t| \geq T_0$, because $a_1 > a_2$. Because $H(b,t)$ is harmonic for $t \neq 0$, we may apply the maximum principle to verify (3.9) for $(b,t) \in \left[-\frac{1}{2},\frac{1}{2}\right] \times ([-T_0,0) \cup (0,T_0])$. We conclude that it suffices to show that $H(b,t) \leq 0$ on the boundary of this area, in order to establish the desired result. First, consider the area with $t > 0$. By our choice of T_0 we already know that $H(b,T_0) \leq 0$ for all $b \in \left[-\frac{1}{2},\frac{1}{2}\right]$. Because $d > \frac{1}{2}$ and $|b| \leq \frac{1}{2}$, we have that $d \pm b > 0$ for all $b \in \left[-\frac{1}{2},\frac{1}{2}\right]$. Thus,

$$\begin{aligned}\lim_{t\to 0^+} f(b,t) = \lim_{t\to 0^+}\Bigg[& 2\arctan\left(\frac{b+d}{t}\right) + 2\arctan\left(\frac{-b+d}{t}\right)\\ & -\arctan\left(\frac{b+2d}{t}\right) - \arctan\left(\frac{-b+2d}{t}\right)\Bigg] = \pi.\end{aligned}$$

And therefore also

$$\begin{aligned}\lim_{t\to 0^+} H(b,t) =& \pi - \frac{\pi}{4}\left[da_1\left(\frac{1}{d+b}+\frac{1}{d-b}\right) + d^2a_2\left(\frac{1}{(d+b)^2}+\frac{1}{(d-b)^2}\right)\right]\\ \leq& \pi - \frac{\pi}{4}\left[da_1\left(\frac{1}{d}+\frac{1}{d}\right) + d^2a_2\left(\frac{1}{d^2}+\frac{1}{d^2}\right)\right]\\ \leq& \pi - \frac{\pi}{4}\left[2a_1+2a_2\right] = 0.\end{aligned}$$

Next, observe that $H(b,t)$ is symmetric about $b=0$. Therefore, it suffices to show that $H\left(\frac{1}{2},t\right) \leq 0$ for all $t > 0$ to verify that $H(b,t) \le 0$ on the boundary of $\left[-\frac{1}{2}, \frac{1}{2}\right] \times (0, T_0]$. Let

$$
\begin{aligned}
h(t) := H\left(\frac{1}{2}, t\right) =& 2\arctan\left(\frac{d+\frac{1}{2}}{t}\right) + 2\arctan\left(\frac{d-\frac{1}{2}}{t}\right) \\
&- \arctan\left(\frac{2d+\frac{1}{2}}{t}\right) - \arctan\left(\frac{2d-\frac{1}{2}}{t}\right) \\
&- \frac{\pi}{4}\left[da_1\left(\frac{d+\frac{1}{2}}{\left(d+\frac{1}{2}\right)^2+t^2} + \frac{d-\frac{1}{2}}{\left(d-\frac{1}{2}\right)^2+t^2}\right)\right. \\
&+ d^2 a_2\left(\frac{\left(d+\frac{1}{2}\right)^2 - t^2}{\left(\left(d+\frac{1}{2}\right)^2+t^2\right)^2} + \frac{\left(d-\frac{1}{2}\right)^2 - t^2}{\left(\left(d-\frac{1}{2}\right)^2+t^2\right)^2}\right) \\
&\left. + a_3\left(\frac{2\left(d+\frac{1}{2}\right)t}{\left(\left(d+\frac{1}{2}\right)^2+t^2\right)^2} + \frac{2\left(d-\frac{1}{2}\right)t}{\left(\left(d-\frac{1}{2}\right)^2+t^2\right)^2}\right)\right].
\end{aligned}
$$

The asymptotic behavior of this equation is already controlled by (3.10). Next, we want to analyze the local maxima of h. For this, consider the derivative with respect to t. We obtain

$$
\begin{aligned}
h'(t) =& \frac{2d+\frac{1}{2}}{(2d+\frac{1}{2})^2+t^2} + \frac{2d-\frac{1}{2}}{(2d-\frac{1}{2})^2+t^2} - \frac{2d+1}{(d+\frac{1}{2})^2+t^2} - \frac{2d-1}{(d-\frac{1}{2})^2+t^2} \\
&+ \frac{\pi}{4}\left[da_1\left(\frac{(2d+1)t}{((d+\frac{1}{2})^2+t^2)^2} + \frac{(2d-1)t}{((d-\frac{1}{2})^2+t^2)^2}\right)\right. \\
&+ d^2 a_2\left(\frac{2t(3(d+\frac{1}{2})^2-t^2)}{((d+\frac{1}{2})^2+t^2)^3} + \frac{2t(3(d-\frac{1}{2})^2-t^2)}{((d-\frac{1}{2})^2+t^2)^3}\right) \\
&\left. - a_3\left(\frac{2(d+\frac{1}{2})((d+\frac{1}{2})^2-3t^2)}{((d+\frac{1}{2})^2+t^2)^3} + \frac{2(d-\frac{1}{2})((d-\frac{1}{2})^2-3t^2)}{((d-\frac{1}{2})^2+t^2)^3}\right)\right].
\end{aligned}
$$

Multiplying out this equation for our values of d, a_1, a_2, and a_3 shows that the zeros of $h'(t)$ correspond to the zeros of a thirteenth-degree polynomial with seven real roots, given approximately by $t_1 \approx -0.791179$, $t_2 \approx -0.345961$, $t_3 \approx 0.052031$, $t_4 \approx 0.909110$, $t_5 \approx 1.335121$, $t_6 \approx 2.749949$, and $t_7 \approx 5.863459$. The zeros t_1, t_3, t_5, and t_7 correspond to local minima of $h(t)$, the remaining extrema correspond to local maxima with values $h(t_2) \approx -0.000192$, $h(t_4) \approx -0.000225$, and $h(t_6) \approx -0.000152$. Hence, $H\left(\frac{1}{2}, t\right) = H\left(-\frac{1}{2}, t\right) = h(t) \le 0$ for all $t \in \mathbb{R}$. Thus, we obtain by the maximum principle that $H(b,t) \le 0$ for $b \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ and $0 < t \le T_0$. For the area with $t < 0$ we obtain $H(b, -T_0) \le 0$ by our choice of T_0. Furthermore, notice that

$$
\begin{aligned}
\lim_{t \to 0^-} H(b,t) &= -\pi - \frac{\pi}{4}\left[da_1\left(\frac{1}{d+b} + \frac{1}{d-b}\right) + d^2 a_2\left(\frac{1}{(d+b)^2} + \frac{1}{(d-b)^2}\right)\right] \\
&\le -\pi - \frac{\pi}{4}\left[da_1\left(\frac{1}{d} + \frac{1}{d}\right) + d^2 a_2\left(\frac{1}{d^2} + \frac{1}{d^2}\right)\right] \\
&\le -\pi - \frac{\pi}{4}\left[2a_1 + 2a_2\right] < 0.
\end{aligned}
$$

And by the maximum principle, we conclude that $H(b,t) \le 0$ for $b \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ and $-T_0 \le t < 0$. This proves the upper bound for $f(b,t)$. Now note that $f(b,t)$ and $\frac{2(d+b)t}{((d+b)^2+t^2)^2}$ are odd functions in t, while $\frac{d+b}{(d+b)^2+t^2}$ and $\frac{(d+b)^2-t^2}{((d+b)^2+t^2)^2}$ are even functions in t. Furthermore, we have established the upper bound for all $|b| \leq \frac{1}{2}$ and $t \neq 0$. Hence, the lower bound follows from the upper bound by substituting $t \mapsto -t$. □

Remark 3.2. We will see that the quality of the overall estimate of Theorem 2.1 depends linearly on the size of the product da_1. All we did in Lemma 3.1 was to show that these particular values for d, a_1, a_2, and a_3 are admissible values for the optimization problem, that is, to find values for d, a_1, a_2, and a_3 such that da_1 is minimal under the condition that $H(b,t) \leq 0$ on the boundary of $\left[-\frac{1}{2}, \frac{1}{2}\right] \times \mathbb{R}$. The condition requires $a_1 > a_2$ and $a_1 + a_2 \geq 2$ to control the behavior of $H(b,t)$ for $|t| \to \infty$ and $|t| \to 0$. We see that the terms associated with a_1 and a_2 are positive, even functions in t and that $f(b,t)$ is an odd function in t that is negative for negative t. Hence, we use the a_3 term to distribute the weight of the estimate more evenly across the positive and negative values for t. Without this wild card, inequality $H(b,t) \leq 0$ is trivially true for all $t < 0$. The specific values of d, a_1, a_2, and a_3 for this lemma were found by strategic testing.

Remark 3.3. Further admissible choices for (d, a_1, a_2, a_3) are given in the table below. We index (i) the tuples to allow for easier cross referencing throughout the paper.

i	d	a_1	a_2	a_3
1	0.778	1.028	0.972	0.553
2	0.922	1.041	0.959	0.708
3	1.065	1.051	0.949	0.882
4	1.203	1.064	0.936	1.034
5	1.332	1.081	0.919	1.136
6	1.461	1.095	0.905	1.247

Table 3: Further admissible choices for (d, a_1, a_2, a_3)

These values were chosen to minimize

$$\frac{1}{\pi}\left(\log(d) + 2\log\left(2d - \frac{1}{2}\right) - 4\log\left(d - \frac{1}{2}\right)\right)$$
$$+\frac{da_1}{2}\left(\frac{1}{d-\frac{1}{2}} - \frac{1}{d}\right) + \frac{\sqrt{(d^2a_2)^2 + a_3^2}}{2}\left(\frac{1}{\left(d-\frac{1}{2}\right)^2} - \frac{1}{2d^2}\right)$$

for $da_1 \leq 0.64 + 0.16i$ by strategic testing. The choice was to minimize this quantity because the dominant contribution to C_2 term in Theorem 2.1 is roughly proportional to this number, while C_1 is linearly dependent on the product da_1.

Now, with these bounds we may estimate the sum in (3.8). Recall that we use $\rho = \beta + i\omega$ and $b = \frac{1}{2} - \beta$.

We obtain the upper bound given by

$$
\begin{aligned}
\frac{1}{\pi}\sum_{\rho} f(b,T-\omega) \leq & \frac{da_1}{4}\operatorname{Re}\left(\sum_{\rho}\frac{1}{\frac{1}{2}+d+iT-\rho}+\frac{1}{\frac{1}{2}+d+iT-(1-\bar{\rho})}\right)\\
&+\frac{d^2a_2}{4}\operatorname{Re}\left(\sum_{\rho}\frac{1}{(\frac{1}{2}+d+iT-\rho)^2}+\frac{1}{(\frac{1}{2}+d+iT-(1-\bar{\rho}))^2}\right)\\
&-\frac{a_3}{4}\operatorname{Im}\left(\sum_{\rho}\frac{1}{(\frac{1}{2}+d+iT-\rho)^2}+\frac{1}{(\frac{1}{2}+d+iT-(1-\bar{\rho}))^2}\right)\\
=&E_2(\xi_K)(T)+E_3(\xi_K)(T)
\end{aligned}
\tag{3.11}
$$

and analogously the lower bound given by

$$
\begin{aligned}
\frac{1}{\pi}\sum_{\rho} f(b,T-\omega) \geq & -\frac{da_1}{4}\operatorname{Re}\left(\sum_{\rho}\frac{1}{\frac{1}{2}+d+iT-\rho}+\frac{1}{\frac{1}{2}+d+iT-(1-\bar{\rho})}\right)\\
&-\frac{d^2a_2}{4}\operatorname{Re}\left(\sum_{\rho}\frac{1}{(\frac{1}{2}+d+iT-\rho)^2}+\frac{1}{(\frac{1}{2}+d+iT-(1-\bar{\rho}))^2}\right)\\
&-\frac{a_3}{4}\operatorname{Im}\left(\sum_{\rho}\frac{1}{(\frac{1}{2}+d+iT-\rho)^2}+\frac{1}{(\frac{1}{2}+d+iT-(1-\bar{\rho}))^2}\right)\\
=&-E_2(\xi_K)(T)+E_3(\xi_K)(T).
\end{aligned}
\tag{3.12}
$$

Here, $E_2(f)(t)$ and $E_3(f)(t)$ are operators given by

$$
\begin{aligned}
E_2(f)(t) :=&\left[\frac{da_1}{2}\operatorname{Re}\left(\frac{f'}{f}\right)-\frac{d^2a_2}{2}\operatorname{Re}\left(\frac{f''}{f}-\left(\frac{f'}{f}\right)^2\right)\right]\left(\frac{1}{2}+d+it\right) \text{ and}\\
E_3(f)(t) :=&-\frac{a_3}{2}\operatorname{Im}\left(\frac{f''}{f}-\left(\frac{f'}{f}\right)^2\right)\left(\frac{1}{2}+d+it\right).
\end{aligned}
$$

Notice that E_1, E_2 and E_3 satisfy $E_i(f\cdot g)(t) = E_i(f)(t)+E_i(g)(t)$. Equations (3.11) and (3.12) follow due to this fact and equations (3.4) and (3.5). We may now decompose $\xi_K(s)$ according to (1.2). We thus obtain for $E_u := E_1+E_2+E_3$ and $E_l := E_1-E_2+E_3$, that

$$
\begin{aligned}
E_u(\xi_K)(T) =&E_u(s(s-1))(T)+E_u\left(d_K^{s/2}\right)(T)+E_u(\gamma_K)(T)+E_u(\zeta_{\mathrm{K}})(T) \text{ and}\\
E_l(\xi_K)(T) =&E_l(s(s-1))(T)+E_l\left(d_K^{s/2}\right)(T)+E_l(\gamma_K)(T)+E_l(\zeta_{\mathrm{K}})(T).
\end{aligned}
\tag{3.13}
$$

By equation (3.6) we obtain that

$$E_l(\xi_K)(T) \leq N_K(T) \leq E_u(\xi_K)(T).$$

Next, we will use equation (3.13) to estimate each term individually.

Lemma 3.4. *For d, a_1, a_2, a_3 as in Lemma 3.1 and $T \geq 1$ we obtain*

$$\begin{aligned} E_u(s(s-1))(T) \leq & 2.001 \quad \textit{and} \\ E_l(s(s-1))(T) \geq & 1.547. \end{aligned}$$

Proof. Plugging in the equation for $E_u(s(s-1))(T)$ yields

$$\begin{aligned} E_u(s(s-1))(T) = & \frac{4}{\pi}\left(\arctan\left(\frac{T}{d+\frac{1}{2}}\right) + \arctan\left(\frac{T}{d-\frac{1}{2}}\right)\right) \\ & - \frac{2}{\pi}\left(\arctan\left(\frac{T}{2d+\frac{1}{2}}\right) + \arctan\left(\frac{T}{2d-\frac{1}{2}}\right)\right) \\ & + \frac{da_1}{2}\left(\frac{d+\frac{1}{2}}{(d+\frac{1}{2})^2+T^2} + \frac{d-\frac{1}{2}}{(d-\frac{1}{2})^2+T^2}\right) \\ & + \frac{d^2a_2}{2}\left(\frac{(d+\frac{1}{2})^2-T^2}{((d+\frac{1}{2})^2+T^2)^2} + \frac{(d-\frac{1}{2})^2-T^2}{((d-\frac{1}{2})^2+T^2)^2}\right) \\ & - \frac{a_3}{2}\left(\frac{2(d+\frac{1}{2})T}{((d+\frac{1}{2})^2+T^2)^2} + \frac{2(d-\frac{1}{2})T}{((d-\frac{1}{2})^2+T^2)^2}\right). \end{aligned}$$

From this expression, we observe that $E_u(s(s-1))(T)$ approaches 2 as $T \to \infty$. Considering the derivative, we observe that $E_u(s(s-1))(T)$ has one local maxima around $T \approx 29.995179$ that is less than 2.001. Hence, $E_u(s(s-1))(T) \leq 2.001$ for $T \geq 1$. For E_l we obtain

$$\begin{aligned} E_l(s(s-1))(T,d) = & \frac{4}{\pi}\left(\arctan\left(\frac{T}{d+\frac{1}{2}}\right) + \arctan\left(\frac{T}{d-\frac{1}{2}}\right)\right) \\ & - \frac{2}{\pi}\left(\arctan\left(\frac{T}{2d+\frac{1}{2}}\right) + \arctan\left(\frac{T}{2d-\frac{1}{2}}\right)\right) \\ & - \frac{da_1}{2}\left(\frac{d+\frac{1}{2}}{(d+\frac{1}{2})^2+T^2} + \frac{d-\frac{1}{2}}{(d-\frac{1}{2})^2+T^2}\right) \\ & - \frac{d^2a_2}{2}\left(\frac{(d+\frac{1}{2})^2-T^2}{((d+\frac{1}{2})^2+T^2)^2} + \frac{(d-\frac{1}{2})^2-T^2}{((d-\frac{1}{2})^2+T^2)^2}\right) \\ & - \frac{a_3}{2}\left(\frac{2(d+\frac{1}{2})T}{((d+\frac{1}{2})^2+T^2)^2} + \frac{2(d-\frac{1}{2})T}{((d-\frac{1}{2})^2+T^2)^2}\right). \end{aligned}$$

We observe that $E_l(s(s-1))(T)$ approaches 2 as $T \to \infty$ and from the derivative we conclude that $E_l(s(s-1))(T)$ has no local extrema for $T \in [0, +\infty)$. From comparing the boundary values we conclude that $E_l(s(s-1))(T) \geq E_l(s(s-1))(1) \geq 1.547$ for $T \geq 1$. □

Next, we estimate the discriminant term.

Lemma 3.5. *For d, a_1, a_2, a_3 as in Lemma 3.1 and $T \geq 1$ we obtain*

$$\begin{aligned} E_u\left(d_K^{s/2}\right)(T) = & \left(\frac{T}{\pi} + \frac{da_1}{4}\right)\log(d_K) \quad \textit{and} \\ E_l\left(d_K^{s/2}\right)(T) = & \left(\frac{T}{\pi} - \frac{da_1}{4}\right)\log(d_K), \end{aligned}$$

where d_K denotes the absolute discriminant of the number field K.

Proof. Plugging in the equations for $E_u\left(d_K^{s/2}\right)(T)$ and $E_l\left(d_K^{s/2}\right)(T)$ yields the result. We have

$$\begin{aligned}
E_u\left(d_K^{s/2}\right)(T) &= \frac{4}{\pi}\operatorname{Im}\log\left(d_K^{\frac{\frac{1}{2}+d+iT}{2}}\right) - \frac{2}{\pi}\operatorname{Im}\log\left(d_K^{\frac{\frac{1}{2}+2d+iT}{2}}\right) + \frac{da_1}{2}\operatorname{Re}\left(\frac{\log(d_K)}{2}\right)\\
&= \left(\frac{T}{\pi} + \frac{da_1}{4}\right)\log(d_K) \quad \text{and}\\
E_l\left(d_K^{s/2}\right)(T) &= \frac{4}{\pi}\operatorname{Im}\log\left(d_K^{\frac{\frac{1}{2}+d+iT}{2}}\right) - \frac{2}{\pi}\operatorname{Im}\log\left(d_K^{\frac{\frac{1}{2}+2d+iT}{2}}\right) - \frac{da_1}{2}\operatorname{Re}\left(\frac{\log(d_K)}{2}\right)\\
&= \left(\frac{T}{\pi} - \frac{da_1}{4}\right)\log(d_K).
\end{aligned}$$

□

Before we estimate the contribution of the gamma term $\gamma_K(s)$, we require some useful estimates for the logarithmic gamma function and its derivatives. The logarithmic gamma function is a special function, in the sense that it is not any particular branch of the logarithm applied to the gamma function, but instead it is defined as the integral $\int_1^s \frac{\Gamma'}{\Gamma}(z)dz$ with a branch cut along the negative real axis. This distinction is important to us because we specifically defined $\operatorname{Im}\log(f)$ as the continuous variation of the argument of the function f stating from the real axis. So, in particular $\operatorname{Im}\log(\Gamma(s))$ will be the imaginary part of the logarithmic gamma function.

Corollary 3.6. *Let $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 0$, $s = \sigma + it$. We then have*

$$\begin{aligned}
\operatorname{Im}\log(\Gamma(s)) &= \frac{t}{2}\log\left(\sigma^2+t^2\right) + \left(\sigma - \frac{1}{2}\right)\arctan\left(\frac{t}{\sigma}\right) - t - \frac{t}{12(\sigma^2+t^2)} + R_1(s),\\
\operatorname{Re}\left(\psi(s)\right) &= \frac{1}{2}\log\left(\sigma^2+t^2\right) - \frac{\sigma}{2(\sigma^2+t^2)} - \frac{\sigma^2-t^2}{12(\sigma^2+t^2)^2} + R_2(s),\\
\operatorname{Re}(\psi_1(s)) &= \frac{\sigma}{\sigma^2+t^2} + \frac{\sigma^2-t^2}{2(\sigma^2+t^2)^2} + R_3(s), \quad \textit{and}\\
\operatorname{Im}(\psi_1(s)) &= \frac{t}{\sigma^2+t^2} - \frac{\sigma t}{(\sigma^2+t^2)^2} + R_4(s).
\end{aligned}$$

Here,

$$\begin{aligned}
|R_1(s)| &\le \frac{1}{360|s|^3} + \frac{1}{1022\sigma|s|^3},\\
|R_2(s)| &\le \frac{1}{120\sigma|s|^3}, \quad \textit{and}\\
|R_3(s)|, |R_4(s)| &\le \frac{1}{6|s|^3} + \frac{1}{23\sigma|s|^3}.
\end{aligned}$$

Proof. These estimates rely on Binet's first formula for the logarithmic gamma function [WW20, Theorem 12.31] which is valid for $\operatorname{Re}(s) > 0$. With this formula for the logarithmic gamma function and the notion that the imaginary part of the logarithmic gamma function is exactly $\operatorname{Im}\log(\Gamma(s))$ as we defined it for equation (3.3) we obtain

$$\operatorname{Im}\log(\Gamma(s)) = \frac{t}{2}\log(\sigma^2+t^2) + \left(\sigma - \frac{1}{2}\right)\arctan\left(\frac{t}{\sigma}\right) - t + \operatorname{Im}\left(\int_0^\infty f(u)e^{-us}du\right), \tag{3.14}$$

where $f(u) = \frac{1}{u}\left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}\right)$. Next, we can observe that $\lim_{u\to 0} f(u) = \frac{1}{12}$, $\lim_{u\to 0} f'(u) = 0$, $\lim_{u\to 0} f''(u) = -\frac{1}{360}$, and $|f'''(u)| \le \frac{1}{1022}$ for $u > 0$. Now, take equation (3.14) and apply partial integration. We thus obtain that

$$\begin{aligned}\int_0^\infty \left[\frac{1}{u}\left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}\right)\right]e^{-us}du = \int_0^\infty f(u)e^{-us}du \\ = \frac{1}{12s} - \frac{1}{360s^3} + \frac{1}{s^3}\int_0^\infty f'''(u)e^{-us}du.\end{aligned}$$

Hence,

$$|R_1(s)| = \left|\operatorname{Im}\left(\int_0^\infty \left[\frac{1}{u}\left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}\right)\right]e^{-us}du - \frac{1}{12s}\right)\right| \le \frac{1}{360|s|^3} + \frac{1}{1022\sigma|s|^3}.$$

Considering the derivative of Binet's first formula for the logarithmic gamma function yields that

$$\operatorname{Re}(\psi(s)) = \frac{1}{2}\log(\sigma^2+t^2) - \frac{\sigma}{2(\sigma^2+t^2)} - \operatorname{Re}\left(\int_0^\infty g(u)e^{-us}du\right), \tag{3.15}$$

where $g(u) = \frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}$. We then have $\lim_{u\to 0} g(u) = 0$, $\lim_{u\to 0} g'(u) = \frac{1}{12}$, $\lim_{u\to 0} g''(u) = 0$ and $|g'''(u)| \le \frac{1}{120}$ for $u > 0$. Now, if we take equation (3.15), we obtain

$$-\int_0^\infty \left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}\right)e^{-us}du = -\int_0^\infty g(u)e^{-us}du = -\frac{1}{12s^2} - \frac{1}{s^3}\int_0^\infty g'''(u)e^{-us}du.$$

Hence,

$$|R_2(s)| \le \left|\operatorname{Re}\left(\int_0^\infty \left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}\right)e^{-us}du - \frac{1}{12s^2}\right)\right| \le \frac{1}{120\sigma|s|^3}.$$

Lastly, consider the second derivative of Binet's first formula for the logarithmic gamma function. From this we obtain that

$$\begin{aligned}\operatorname{Re}(\psi_1(s)) &= \frac{\sigma}{\sigma^2+t^2} + \frac{\sigma^2-t^2}{2(\sigma^2+t^2)^2} + \operatorname{Re}\left(\int_0^\infty h(u)e^{-us}du\right), \text{ and} \\ \operatorname{Im}(\psi_1(s)) &= \frac{t}{\sigma^2+t^2} - \frac{\sigma t}{(\sigma^2+t^2)^2} + \operatorname{Im}\left(\int_0^\infty h(u)e^{-us}du\right),\end{aligned} \tag{3.16}$$

where $h(u) = u\left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}\right)$. Then $\lim_{u\to 0} h(u) = 0$, $\lim_{u\to 0} h'(u) = 0$, $\lim_{u\to 0} h''(u) = \frac{1}{6}$ and $|h'''(u)| \le \frac{1}{23}$ for $u > 0$. Therefore,

$$\int_0^\infty \left[u\left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u-1}\right)\right]e^{-us}du = \int_0^\infty h(u)e^{-us}du = \frac{1}{6s^3} + \frac{1}{s^3}\int_0^\infty h'''(u)e^{-us}du.$$

And in sight of (3.16),

$$|R_3(s)| \le \left|\int_0^\infty \left[u\left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u - 1}\right)\right] e^{-us} du\right| \le \frac{1}{6|s|^3} + \frac{1}{23\sigma|s|^3} \quad \text{and}$$
$$|R_4(s)| \le \left|\int_0^\infty \left[u\left(\frac{1}{2} - \frac{1}{u} + \frac{1}{e^u - 1}\right)\right] e^{-us} du\right| \le \frac{1}{6|s|^3} + \frac{1}{23\sigma|s|^3}.$$

□

Now we can estimate the contribution of the gamma factor $\gamma_K(s)$.

Lemma 3.7. *For d, a_1, a_2, a_3 as in Lemma 3.1 and $T \ge 1$ we obtain*

$$E_u(\gamma_K)(T) \le \frac{n_K T}{\pi} \log\left(\frac{T}{2\pi e}\right) + \frac{n_K d a_1}{4} \log\left(\frac{T}{2\pi}\right) + 0.092 n_K \quad \textit{and}$$
$$E_l(\gamma_K)(T) \ge \frac{n_K T}{\pi} \log\left(\frac{T}{2\pi e}\right) - \frac{n_K d a_1}{4} \log\left(\frac{T}{2\pi}\right) - 0.25 n_K.$$

Proof. First, use (1.4) to decompose the gamma factor into its r_1- and r_2-dependent parts. We first estimate the γ_1 term. Plugging in the equation yields

$$\begin{aligned}
E_u(\gamma_1)(T) = &\frac{4}{\pi} \operatorname{Im} \log\left(\pi^{-(\frac{1}{2}+d+iT)/2} \Gamma\left(\frac{\frac{1}{2} + d + iT}{2}\right)\right) \\
&- \frac{2}{\pi} \operatorname{Im} \log\left(\pi^{-(\frac{1}{2}+2d+iT)/2} \Gamma\left(\frac{\frac{1}{2} + 2d + iT}{2}\right)\right) \\
&+ \frac{d a_1}{2} \operatorname{Re}\left(-\frac{\log(\pi)}{2} + \frac{1}{2}\psi\left(\frac{\frac{1}{2} + d + iT}{2}\right)\right) \\
&- \frac{d^2 a_2}{2} \operatorname{Re}\left(\frac{1}{4}\psi_1\left(\frac{\frac{1}{2} + d + iT}{2}\right)\right) - \frac{a_3}{2} \operatorname{Im}\left(\frac{1}{4}\psi_1\left(\frac{\frac{1}{2} + d + iT}{2}\right)\right).
\end{aligned}$$

The exponential terms equal $-\left(\frac{T}{\pi} + \frac{d a_1}{4}\right)\log(\pi)$ and by Corollary 3.6 we obtain that

$$E_u(\gamma_1)(T) = \frac{T}{\pi} \log\left(\frac{T}{2\pi e}\right) + \frac{d a_1}{4} \log\left(\frac{T}{2\pi}\right) + U_{1,1}(T) + U_{1,2}(T),$$

where

$$\begin{aligned}U_{1,1}(T) =&\frac{T}{\pi}\log\left(1+\left(\frac{1+2d}{2T}\right)^2\right)-\frac{T}{2\pi}\log\left(1+\left(\frac{1+4d}{2T}\right)^2\right)\\&+\frac{da_1}{8}\log\left(1+\left(\frac{1+2d}{2T}\right)^2\right)+\frac{2d-1}{\pi}\arctan\left(\frac{T}{\frac{1}{2}+d}\right)\\&-\frac{4d-1}{2\pi}\arctan\left(\frac{T}{\frac{1}{2}+2d}\right)-\frac{2T}{3\pi((\frac{1}{2}+d)^2+T^2)}\\&+\frac{T}{3\pi((\frac{1}{2}+2d)^2+T^2)}-\frac{da_1}{4}\left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2}+\frac{(\frac{1}{2}+d)^2-T^2}{3((\frac{1}{2}+d)^2+T^2)^2}\right)\\&-\frac{d^2a_2}{4}\left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2}+\frac{(\frac{1}{2}+d)^2-T^2}{((\frac{1}{2}+d)^2+T^2)^2}\right)\\&-\frac{a_3}{4}\left(\frac{T}{(\frac{1}{2}+d)^2+T^2)}-\frac{2(\frac{1}{2}+d)T}{((\frac{1}{2}+d)^2+T^2)^2}\right),\end{aligned}$$

and

$$\begin{aligned}U_{1,2}(T) =&\frac{4}{\pi}R_1\left(\frac{1}{4}+\frac{d}{2}+i\frac{T}{2}\right)-\frac{2}{\pi}R_1\left(\frac{1}{4}+d+i\frac{T}{2}\right)+\frac{da_1}{4}R_2\left(\frac{1}{4}+\frac{d}{2}+i\frac{T}{2}\right)\\&-\frac{d^2a_2}{8}R_3\left(\frac{1}{4}+\frac{d}{2}+i\frac{T}{2}\right)-\frac{a_3}{8}R_4\left(\frac{1}{4}+\frac{d}{2}+i\frac{T}{2}\right).\end{aligned}$$

Invoking Corollary 3.6 yields that

$$\begin{aligned}U_{1,2}(T) \leq&\frac{4}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{4}+\frac{d}{2})}\right)\frac{1}{((\frac{1}{4}+\frac{d}{2})^2+\frac{T^2}{4})^{\frac{3}{2}}}\\&+\frac{2}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{4}+d)}\right)\frac{1}{((\frac{1}{4}+d)^2+\frac{T^2}{4})^{\frac{3}{2}}}\\&+\frac{da_1}{4}\frac{1}{120(\frac{1}{4}+\frac{d}{2})((\frac{1}{4}+\frac{d}{2})^2+\frac{T^2}{4})^{\frac{3}{2}}}\\&+\left(\frac{d^2a_2}{8}+\frac{a_3}{8}\right)\left(\frac{1}{6}+\frac{1}{23(\frac{1}{4}+\frac{d}{2})}\right)\frac{1}{((\frac{1}{4}+\frac{d}{2})^2+\frac{T^2}{4})^{\frac{3}{2}}}\\&=:UB_1(T).\end{aligned}$$

From considering the derivative of $U_{1,1}(T)+UB_1(T)$ we obtain that the function decreases in the interval $[1,\infty)$ with a maximum at 1. This yields that

$$E_u(\gamma_1)(T) \leq\frac{T}{\pi}\log\left(\frac{T}{2\pi e}\right)+\frac{da_1}{4}\log\left(\frac{T}{2\pi}\right)-0.078$$

for $T \geq 1$. Similarly, we obtain

$$\begin{aligned}
E_l(\gamma_1)(T) =& \frac{4}{\pi} \operatorname{Im} \log \left(\pi^{-(\frac{1}{2}+d+iT)/2} \Gamma \left(\frac{\frac{1}{2}+d+iT}{2} \right) \right) \\
&- \frac{2}{\pi} \operatorname{Im} \log \left(\pi^{-(\frac{1}{2}+2d+iT)/2} \Gamma \left(\frac{\frac{1}{2}+2d+iT}{2} \right) \right) \\
&- \frac{da_1}{2} \operatorname{Re} \left(-\frac{\log(\pi)}{2} + \frac{1}{2} \psi \left(\frac{\frac{1}{2}+d+iT}{2} \right) \right) \\
&+ \frac{d^2 a_2}{2} \operatorname{Re} \left(\frac{1}{4} \psi_1 \left(\frac{\frac{1}{2}+d+iT}{2} \right) \right) - \frac{a_3}{2} \operatorname{Im} \left(\frac{1}{4} \psi_1 \left(\frac{\frac{1}{2}+d+iT}{2} \right) \right),
\end{aligned}$$

and

$$E_l(\gamma_1)(T) = \frac{T}{\pi} \log \left(\frac{T}{2\pi} \right) - \frac{da_1}{4} \log \left(\frac{T}{2\pi} \right) + L_{1,1}(T) + L_{1,2}(T),$$

where

$$\begin{aligned}
L_{1,1}(T) =& \frac{T}{\pi} \log \left(1 + \left(\frac{1+2d}{2T} \right)^2 \right) - \frac{T}{2\pi} \log \left(1 + \left(\frac{1+4d}{2T} \right)^2 \right) \\
&- \frac{da_1}{8} \log \left(1 + \left(\frac{1+2d}{2T} \right)^2 \right) + \frac{2d-1}{\pi} \arctan \left(\frac{T}{\frac{1}{2}+d} \right) \\
&- \frac{4d-1}{2\pi} \arctan \left(\frac{T}{\frac{1}{2}+2d} \right) - \frac{2T}{3\pi((\frac{1}{2}+d)^2+T^2)} \\
&+ \frac{T}{3\pi((\frac{1}{2}+2d)^2+T^2)} + \frac{da_1}{4} \left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2} + \frac{(\frac{1}{2}+d)^2-T^2}{3((\frac{1}{2}+d)^2+T^2)^2} \right) \\
&+ \frac{d^2 a_2}{4} \left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2} + \frac{(\frac{1}{2}+d)^2-T^2}{((\frac{1}{2}+d)^2+T^2)^2} \right) \\
&- \frac{a_3}{4} \left(\frac{T}{(\frac{1}{2}+d)^2+T^2)} - \frac{2(\frac{1}{2}+d)T}{((\frac{1}{2}+d)^2+T^2)^2} \right),
\end{aligned}$$

and

$$\begin{aligned}
L_{1,2}(T) =& \frac{4}{\pi} R_1 \left(\frac{1}{4} + \frac{d}{2} + i\frac{T}{2} \right) - \frac{2}{\pi} R_1 \left(\frac{1}{4} + d + i\frac{T}{2} \right) - \frac{da_1}{4} R_2 \left(\frac{1}{4} + \frac{d}{2} + i\frac{T}{2} \right) \\
&+ \frac{d^2 a_2}{8} R_3 \left(\frac{1}{4} + \frac{d}{2} + i\frac{T}{2} \right) - \frac{a_3}{8} R_4 \left(\frac{1}{4} + \frac{d}{2} + i\frac{T}{2} \right).
\end{aligned}$$

Invoking Corollary 3.6 yields that

$$\begin{aligned}
L_{1,2}(T) \geq & -\frac{4}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{4}+\frac{d}{2})}\right)\frac{1}{((\frac{1}{4}+\frac{d}{2})^2+\frac{T^2}{4})^{\frac{3}{2}}} \\
& -\frac{2}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{4}+d)}\right)\frac{1}{((\frac{1}{4}+d)^2+\frac{T^2}{4})^{\frac{3}{2}}} \\
& -\frac{da_1}{4}\frac{1}{120(\frac{1}{4}+\frac{d}{2})((\frac{1}{4}+\frac{d}{2})^2+\frac{T^2}{4})^{\frac{3}{2}}} \\
& -\left(\frac{d^2a_2}{8}+\frac{a_3}{8}\right)\left(\frac{1}{6}+\frac{1}{23(\frac{1}{4}+\frac{d}{2})}\right)\frac{1}{((\frac{1}{4}+\frac{d}{2})^2+\frac{T^2}{4})^{\frac{3}{2}}} \\
= & -UB_1(T) =: LB_1(T).
\end{aligned}$$

We conclude that $L_{1,1}(T) + L_{1,2}(T) \geq L_{1,1}(T) + LB_1(T)$. Considering the derivative for this function shows that it is decreasing in the interval $[1, \infty)$ and we obtain that

$$L_{1,1}(T) + L_{1,2}(T) \geq \lim_{t\to\infty} L_{1,1}(t) + LB_1(t) = -\frac{1}{4}.$$

We deduce that

$$E_l(\gamma_1)(T) \geq \frac{T}{\pi}\log\left(\frac{T}{2\pi e}\right) - \frac{da_1}{4}\log\left(\frac{T}{2\pi}\right) - 0.25$$

for $T \geq 1$. Next, consider the γ_2 terms. Plugging in the equation yields

$$\begin{aligned}
E_u(\gamma_2)(T) = & \frac{4}{\pi}\operatorname{Im}\log\left(\pi^{-(\frac{1}{2}+d+iT)}2^{1-(\frac{1}{2}+d+iT)}\Gamma\left(\frac{1}{2}+d+iT\right)\right) \\
& -\frac{2}{\pi}\operatorname{Im}\log\left(\pi^{-(\frac{1}{2}+2d+iT)}2^{1-(\frac{1}{2}+2d+iT)}\Gamma\left(\frac{1}{2}+2d+iT\right)\right) \\
& +\frac{da_1}{2}\operatorname{Re}\left(-\log(2\pi)+\psi\left(\frac{1}{2}+d+iT\right)\right) \\
& -\frac{d^2a_2}{2}\operatorname{Re}\left(\psi_1\left(\frac{1}{2}+d+iT\right)\right)-\frac{a_3}{2}\operatorname{Im}\left(\psi_1\left(\frac{1}{2}+d+iT\right)\right).
\end{aligned}$$

The contribution of the exponential term is $-\left(\frac{2T}{\pi}+\frac{da_1}{2}\right)\log(2\pi)$ and by Corollary 3.6 we obtain that

$$E_u(\gamma_2)(T) = \frac{2T}{\pi}\log\left(\frac{T}{2\pi e}\right) + \frac{da_1}{2}\log\left(\frac{T}{2\pi}\right) + U_{2,1}(T) + U_{2,2}(T),$$

where

$$
\begin{aligned}
U_{2,1}(T) =& \frac{2T}{\pi}\log\left(1+\left(\frac{1+2d}{2T}\right)^2\right) - \frac{T}{\pi}\log\left(1+\left(\frac{1+4d}{2T}\right)^2\right) \\
&+ \frac{da_1}{4}\log\left(1+\left(\frac{1+2d}{2T}\right)^2\right) + \frac{4d}{\pi}\arctan\left(\frac{T}{\frac{1}{2}+d}\right) \\
&- \frac{4d}{\pi}\arctan\left(\frac{T}{\frac{1}{2}+2d}\right) - \frac{T}{3\pi((\frac{1}{2}+d)^2+T^2)} \\
&+ \frac{T}{6\pi((\frac{1}{2}+2d)^2+T^2)} - \frac{da_1}{4}\left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2} + \frac{(\frac{1}{2}+d)^2-T^2}{6((\frac{1}{2}+d)^2+T^2)^2}\right) \\
&- \frac{d^2a_2}{2}\left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2} + \frac{(\frac{1}{2}+d)^2-T^2}{2((\frac{1}{2}+d)^2+T^2)^2}\right) \\
&- \frac{a_3}{2}\left(\frac{T}{(\frac{1}{2}+d)^2+T^2)} - \frac{(\frac{1}{2}+d)T}{((\frac{1}{2}+d)^2+T^2)^2}\right),
\end{aligned}
$$

and

$$
\begin{aligned}
U_{2,2}(T) =& \frac{4}{\pi}R_1\left(\frac{1}{2}+d+iT\right) - \frac{2}{\pi}R_1\left(\frac{1}{2}+2d+iT\right) + \frac{da_1}{2}R_2\left(\frac{1}{2}+d+iT\right) \\
&- \frac{d^2a_2}{2}R_3\left(\frac{1}{2}+d+iT\right) - \frac{a_3}{2}R_4\left(\frac{1}{2}+d+iT\right).
\end{aligned}
$$

Invoking Corollary 3.6 yields that

$$
\begin{aligned}
U_{2,2}(T) \leq& \frac{4}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{2}+d)}\right)\frac{1}{((\frac{1}{2}+d)^2+T^2)^{\frac{3}{2}}} \\
&+ \frac{2}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{2}+2d)}\right)\frac{1}{((\frac{1}{2}+2d)^2+T^2)^{\frac{3}{2}}} \\
&+ \frac{da_1}{2}\frac{1}{120(\frac{1}{2}+d)((\frac{1}{2}+d)^2+T^2)^{\frac{3}{2}}} \\
&+ \left(\frac{d^2a_2}{2}+\frac{a_3}{2}\right)\left(\frac{1}{6}+\frac{1}{23(\frac{1}{2}+d)}\right)\frac{1}{((\frac{1}{2}+d)^2+T^2)^{\frac{3}{2}}} \\
=:& UB_2(T).
\end{aligned}
$$

Then the function $U_{2,1}(T)+UB_2(T)$ is decreasing on the interval $[1,\infty)$ with a maxima at 1 yielding that $U_{2,1}(T)+U_{2,2}(T) \leq U_{2,1}(T)+UB_2(T) \leq U_{2,1}(1)+UB_2(1) \leq 0.184$ for $T \geq 1$. Hence,

$$
E_u(\gamma_2)(T) \leq \frac{2T}{\pi}\log\left(\frac{T}{2\pi e}\right) + \frac{da_1}{2}\log\left(\frac{T}{2\pi}\right) + 0.184
$$

for $T \geq 1$. Plugging γ_2 into the equation for the lower estimate yields

$$\begin{aligned}
E_l(\gamma_2)(T) =& \frac{4}{\pi} \operatorname{Im} \log \left(\pi^{-\left(\frac{1}{2}+d+iT\right)} 2^{1-\left(\frac{1}{2}+d+iT\right)} \Gamma\left(\frac{1}{2}+d+iT\right)\right) \\
&- \frac{2}{\pi} \operatorname{Im} \log \left(\pi^{-\left(\frac{1}{2}+2d+iT\right)} 2^{1-\left(\frac{1}{2}+2d+iT\right)} \Gamma\left(\frac{1}{2}+2d+iT\right)\right) \\
&- \frac{da_1}{2} \operatorname{Re}\left(-\log(2\pi) + \psi\left(\frac{1}{2}+d+iT\right)\right) \\
&+ \frac{d^2 a_2}{2} \operatorname{Re}\left(\psi_1\left(\frac{1}{2}+d+iT\right)\right) - \frac{a_3}{2} \operatorname{Im}\left(\psi_1\left(\frac{1}{2}+d+iT\right)\right).
\end{aligned}$$

The contribution of the exponential term is $-\left(\frac{2T}{\pi} - \frac{da_1}{2}\right)\log(2\pi)$ and by Corollary 3.6 we obtain that

$$E_l(\gamma_2)(T) = \frac{2T}{\pi} \log\left(\frac{T}{2\pi e}\right) - \frac{da_1}{2} \log\left(\frac{T}{2\pi}\right) + L_{2,1}(T) + L_{2,2}(T),$$

where

$$\begin{aligned}
L_{2,1}(T) =& \frac{2T}{\pi} \log\left(1 + \left(\frac{1+2d}{2T}\right)^2\right) - \frac{T}{\pi} \log\left(1 + \left(\frac{1+4d}{2T}\right)^2\right) \\
&- \frac{da_1}{4} \log\left(1 + \left(\frac{1+2d}{2T}\right)^2\right) + \frac{4d}{\pi} \arctan\left(\frac{T}{\frac{1}{2}+d}\right) - \frac{4d}{\pi} \arctan\left(\frac{T}{\frac{1}{2}+2d}\right) \\
&- \frac{T}{3\pi((\frac{1}{2}+d)^2+T^2)} + \frac{T}{6\pi((\frac{1}{2}+2d)^2+T^2)} \\
&+ \frac{da_1}{4}\left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2} + \frac{(\frac{1}{2}+d)^2-T^2}{6((\frac{1}{2}+d)^2+T^2)^2}\right) \\
&+ \frac{d^2a_2}{2}\left(\frac{\frac{1}{2}+d}{(\frac{1}{2}+d)^2+T^2} + \frac{(\frac{1}{2}+d)^2-T^2}{2((\frac{1}{2}+d)^2+T^2)^2}\right) \\
&- \frac{a_3}{2}\left(\frac{T}{(\frac{1}{2}+d)^2+T^2)} - \frac{(\frac{1}{2}+d)T}{((\frac{1}{2}+d)^2+T^2)^2}\right),
\end{aligned}$$

and

$$\begin{aligned}
L_{2,2}(T) =& \frac{4}{\pi} R_1\left(\frac{1}{2}+d+iT\right) - \frac{2}{\pi} R_1\left(\frac{1}{2}+2d+iT\right) - \frac{da_1}{2} R_2\left(\frac{1}{2}+d+iT\right) \\
&+ \frac{d^2a_2}{2} R_3\left(\frac{1}{2}+d+iT\right) - \frac{a_3}{2} R_4\left(\frac{1}{2}+d+iT\right).
\end{aligned}$$

Invoking Corollary 3.6 yields that

$$\begin{aligned}
L_{2,2}(T) \geq & -\frac{4}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{2}+d)}\right)\frac{1}{((\frac{1}{2}+d)^2+T^2)^{\frac{3}{2}}} \\
& -\frac{2}{\pi}\left(\frac{1}{360}+\frac{1}{1022(\frac{1}{2}+2d)}\right)\frac{1}{((\frac{1}{2}+2d)^2+T^2)^{\frac{3}{2}}} \\
& -\frac{da_1}{2}\frac{1}{120(\frac{1}{2}+d)((\frac{1}{2}+d)^2+T^2)^{\frac{3}{2}}} \\
& -\left(\frac{d^2a_2}{2}+\frac{a_3}{2}\right)\left(\frac{1}{6}+\frac{1}{23(\frac{1}{2}+d)}\right)\frac{1}{((\frac{1}{2}+d)^2+T^2)^{\frac{3}{2}}} \\
= & -UB_2(T) =: LB_2(T).
\end{aligned}$$

Then $L_{2,1}(T)+L_{2,2}(T) \geq L_{2,1}+LB_2(T)$. This function decreases in the interval $[1,\infty)$, and hence

$$L_{2,1}(T)+L_{2,2}(T) \geq \lim_{t\to\infty} L_{2,1}(t)+LB_2(t) = 0$$

for $T \geq 1$. Altogether, we then obtain

$$E_l(\gamma_2)(T) \geq \frac{2T}{\pi}\log\left(\frac{T}{2\pi e}\right)-\frac{da_1}{2}\log\left(\frac{T}{2\pi}\right)$$

for $T \geq 1$. Collecting the estimate for γ_1 and γ_2 gives

$$\begin{aligned}
E_u(\gamma_K)(T) \leq & \frac{n_K T}{\pi}\log\left(\frac{T}{2\pi e}\right)+\frac{n_K d a_1}{4}\log\left(\frac{T}{2\pi}\right)+0.092n_K \quad \text{and} \\
E_l(\gamma_K(T) \geq & \frac{n_K T}{\pi}\log\left(\frac{T}{2\pi e}\right)-\frac{n_K d a_1}{4}\log\left(\frac{T}{2\pi}\right)-0.25n_K
\end{aligned}$$

for $T \geq 1$. Here we use that $n_K = r_1 + 2r_2$. □

Lastly, we need to estimate the contribution of the zeta terms.

Lemma 3.8. *For d, a_1, a_2, a_3 as in Lemma 3.1 and $T \geq 1$ we obtain*

$$\begin{aligned}
E_u(\zeta_{\mathrm{K}})(T) \leq & 7.911n_K \quad \textit{and} \\
E_l(\zeta_{\mathrm{K}})(T) \geq & -7.911n_K.
\end{aligned}$$

Proof. Because $\frac{1}{2}+d>1$ and $\frac{1}{2}+2d>1$ we may express $\zeta_{\mathrm{K}}(\frac{1}{2}+d), \zeta_{\mathrm{K}}(\frac{1}{2}+2d)$ and related functions with the help of the Euler product (1.6). We obtain that

$$\begin{aligned}
\log\zeta_{\mathrm{K}}(s) = & -\sum_{\mathfrak{p}\subset\mathcal{O}_K}\log(1-(N\mathfrak{p})^{-s}), \\
\frac{\zeta_{\mathrm{K}}'}{\zeta_{\mathrm{K}}}(s) = & \sum_{\mathfrak{p}\subset\mathcal{O}_K}\frac{\log(N\mathfrak{p})}{1-(N\mathfrak{p})^s}, \text{ and} \\
\left(\frac{\zeta_{\mathrm{K}}''}{\zeta_{\mathrm{K}}}-\left(\frac{\zeta_{\mathrm{K}}'}{\zeta_{\mathrm{K}}}\right)^2\right)(s) = & \sum_{\mathfrak{p}\subset\mathcal{O}_K}\frac{(\log(N\mathfrak{p}))^2(N\mathfrak{p})^s}{(1-(N\mathfrak{p})^s)^2}.
\end{aligned} \tag{3.17}$$

Here, $\log \zeta_{\mathrm{K}}(s)$ is again a special function and should be understood as the integral $\int_2^s \frac{\zeta_{\mathrm{K}}'}{\zeta_{\mathrm{K}}}(z) + \log(\zeta_{\mathrm{K}}(2))$, rather than any particular branch of the logarithm applied to $\zeta_{\mathrm{K}}(s)$. We use the notation $\log \zeta_{\mathrm{K}}(s)$ without parentheses to highlight this distinction. We can use these expressions to compute E_u applied to ζ_{K}. We obtain

$$\begin{aligned}
E_u(\zeta_{\mathrm{K}})(T) = \sum_{\mathfrak{p}\subset\mathcal{O}_K} & \frac{4}{\pi}\arctan\left(\frac{\sin\varphi}{\alpha^{\frac{1}{2}+d}-\cos\varphi}\right) - \frac{2}{\pi}\arctan\left(\frac{\sin\varphi}{\alpha^{\frac{1}{2}+2d}-\cos\varphi}\right) \\
&+ \frac{da_1}{2}\log(\alpha)\frac{1-\alpha^{\frac{1}{2}+d}\cos\varphi}{1-2\alpha^{\frac{1}{2}+d}\cos\varphi+\alpha^{1+2d}} \\
&- \frac{d^2a_2}{2}(\log(\alpha))^2\frac{\alpha^{\frac{1}{2}+d}((1+\alpha^{1+2d})\cos\varphi - 2\alpha^{\frac{1}{2}+d})}{(1-2\alpha^{\frac{1}{2}+d}\cos\varphi+\alpha^{1+2d})^2} \\
&- \frac{a_3}{2}(\log(\alpha))^2\frac{\alpha^{\frac{1}{2}+d}(1-\alpha^{1+2d})\sin\varphi}{(1-2\alpha^{\frac{1}{2}+d}\cos(\varphi)+\alpha^{1+2d})^2} \\
= \sum_{\mathfrak{p}\subset\mathcal{O}_K} q_1(\alpha,\varphi) \le & \sum_{\mathfrak{p}\subset\mathcal{O}_K} \max_{\varphi} q_1(\alpha,\varphi),
\end{aligned}$$

where $\alpha := N\mathfrak{p}$ and $\varphi := T\log(N\mathfrak{p})$. Next, we want to show that

$$\max_{\varphi}(q_1(\alpha^m,\varphi)) \le \max_{\varphi}(q_1(\alpha,\varphi)) \tag{3.18}$$

holds for all integers $\alpha \ge 2$ and positive integers m. With this inequality and an application of the fundamental formula for the splitting behavior of primes in a field extension $K/\mathbb{Q}$ we may replace the sum over the prime ideals of $\mathcal{O}_K$ with a sum involving the prime numbers. We first confirm the validity of equation (3.18) numerically for $x \le 100$ with python. To verify the equation for larger x, we first introduce the following notation, write

$$\begin{aligned}
q_1(\alpha^m,\varphi) =& \frac{2}{\pi}f_1(\alpha^m,\varphi) + \frac{da_1}{2}f_2(\alpha^m,\varphi) + \frac{d^2a_2}{2}f_3(\alpha^m,\varphi) + \frac{a_3}{2}f_4(\alpha^m,\varphi), \quad \text{where} \\
f_1(x,\varphi) :=& 2\arctan\left(\frac{\sin\varphi}{x^{\frac{1}{2}+d}-\cos\varphi}\right) - \arctan\left(\frac{\sin\varphi}{x^{\frac{1}{2}+2d}-\cos\varphi}\right), \\
f_2(x,\varphi) :=& \log(x)\,\frac{1-x^{\frac{1}{2}+d}\cos\varphi}{1-2x^{\frac{1}{2}+d}\cos\varphi+x^{1+2d}}, \\
f_3(x,\varphi) :=& -(\log(x))^2\,\frac{x^{\frac{1}{2}+d}\left(\left(1+x^{1+2d}\right)\cos\varphi-2x^{\frac{1}{2}+d}\right)}{\left(1-2x^{\frac{1}{2}+d}\cos\varphi+x^{1+2d}\right)^2}, \quad \text{and} \\
f_4(x,\varphi) :=& -(\log(x))^2\,\frac{x^{\frac{1}{2}+d}\left(1-x^{1+2d}\right)\sin\varphi}{\left(1-2x^{\frac{1}{2}+d}\cos\varphi+x^{1+2d}\right)^2}.
\end{aligned} \tag{3.19}$$

Now, we want to show that $q_1(x,\varphi)$ assumes its maximum with respect to φ in the interval $\varphi \in \left[\frac{\pi}{2},\pi\right]$ if $x \ge 100$. We can then consider the derivative of q_1 with respect to x when $\varphi \in \left[\frac{\pi}{2},\pi\right]$. Notice first that to maximize q_1 we need $\sin(\varphi)$ to be positive, so we can deduce that if q_1 is maximized w.r.t. φ, then $0 \le \varphi \le \pi$. Next, we show that q_1 is increasing w.r.t. φ if $\varphi \in \left[0,\frac{\pi}{2}\right]$ from which we will obtain that q_1

indeed assumes its maximum w.r.t. φ, if $\varphi \in \left[\frac{\pi}{2}, \pi\right]$. To see this, we need to consider the derivatives of the functions f_i with respect to φ. For f_1 we have that

$$\begin{aligned}\frac{\partial}{\partial\varphi} f_1(x,\varphi) =& \frac{2(x^{\frac{1}{2}+d}\cos(\varphi)-1)}{1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d}} - \frac{x^{\frac{1}{2}+2d}\cos(\varphi)-1}{1-2x^{\frac{1}{2}+2d}\cos(\varphi)+x^{1+4d}} \\ =& \frac{(2x^{\frac{3}{2}+5d}-x^{\frac{3}{2}+4d}-2x^{1+3d}\cos(\varphi)+3x^{\frac{1}{2}+2d})\cos(\varphi)-2x^{1+4d}+x^{1+2d}-1}{(1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d})(1-2x^{\frac{1}{2}+2d}\cos(\varphi)+x^{1+4d})} \\ \geq& -\frac{2x^{1+4d}-x^{1+2d}+1}{(1-x^{\frac{1}{2}+d})^2(1-x^{\frac{1}{2}+2d})^2}\end{aligned} \tag{3.20}$$

for $\varphi \in \left[0, \frac{\pi}{2}\right]$ and $x \geq 100$. Furthermore, we can see that

$$\begin{aligned}\frac{\partial}{\partial\varphi} f_1(x,\varphi) \geq& \frac{(2x^{\frac{3}{2}+5d}-x^{\frac{3}{2}+4d}-2x^{1+3d}+3x^{\frac{1}{2}+2d})\frac{1}{\sqrt{2}}-2x^{1+4d}+x^{1+2d}-1}{(1+x^{1+2d})(1+x^{1+4d})} \\ \geq& \frac{(\sqrt{2}x^d-2-\frac{1}{\sqrt{2}}-3)x^{\frac{3}{2}+4d}}{(1+x^{1+2d})(1+x^{1+4d})} \\ \geq& 0\end{aligned} \tag{3.21}$$

for $\varphi \in \left[0, \frac{\pi}{4}\right]$, $x \geq 100$ and $d > 0.5$. Similarly, we can see that

$$\frac{\partial}{\partial\varphi} f_2(x,\varphi) = \frac{x^{\frac{1}{2}+d}\left(x^{1+2d}-1\right)\log(x)\sin(\varphi)}{(1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d})^2} \geq \frac{x^{\frac{1}{2}+d}\left(x^{1+2d}-1\right)\log(x)}{\sqrt{2}(1+x^{1+2d})^2} \tag{3.22}$$

for $\frac{\pi}{4} \leq \varphi \leq \frac{\pi}{2}$ and that $\frac{\partial}{\partial\varphi} f_2(x,\varphi) \geq 0$ if $0 \leq \varphi \leq \frac{\pi}{2}$. Furthermore, notice that

$$\begin{aligned}\frac{\partial}{\partial\varphi}\frac{da_1}{2} f_2(x,\varphi) \geq& \frac{da_1}{2}\frac{x^{\frac{1}{2}+d}\left(x^{1+2d}-1\right)\log(x)}{\sqrt{2}(1+x^{1+2d})^2} \geq \frac{3\log(100)}{64\sqrt{2}} x^{-(\frac{1}{2}+d)} \\ \geq& \frac{3\log(100)100^{1.2}}{64\sqrt{2}} x^{-(1+2d)} \geq 38x^{-(1+2d)}\end{aligned}$$

for $\varphi \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ and $x \geq 100$. Here we use that $da_1 \geq 0.75$ and $d \geq 0.7$. Similarly, we may obtain that

$$\frac{\partial}{\partial\varphi}\frac{2}{\pi} f_2(x,\varphi) \geq -\frac{2}{\pi}\frac{2x^{1+4d}-x^{1+2d}+1}{(1-x^{\frac{1}{2}+d})^2(1-x^{\frac{1}{2}+2d})^2} \geq -\frac{64}{\pi}x^{-(1+2d)} \geq -21x^{-(1+2d)}$$

for $x \geq 100$. From this we obtain that

$$\frac{\partial}{\partial\varphi}\left(\frac{2}{\pi} f_1(x,\varphi) + \frac{da_1}{2} f_2(x,\varphi)\right) \geq 0 \tag{3.23}$$

for $\frac{\pi}{4} \le \varphi \le \frac{\pi}{2}$ and $x \ge 100$. Next, we consider the φ-derivatives of f_3 and f_4. For f_3 we have

$$\begin{aligned}\frac{\partial}{\partial\varphi} f_3(x,\varphi) =& \frac{\log^2(x)\left(\left(2x^{2+4d} + 2x^{1+2d}\right)\cos(\varphi) + x^{\frac{5}{2}+5d} - 6x^{\frac{3}{2}+3d} + x^{\frac{1}{2}+d}\right)\sin(\varphi)}{\left(1 - 2x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)^3} \\ \ge& \frac{\log^2(x)\left(x^{\frac{5}{2}+5d} - 6x^{\frac{3}{2}+3d} + x^{\frac{1}{2}+d}\right)\sin(\varphi)}{\left(1 - 2x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)^3} \\ \ge& \frac{\log^2(x)\left(x^{\frac{5}{2}+5d} - 6x^{\frac{3}{2}+3d} + x^{\frac{1}{2}+d}\right)}{\sqrt{2}\left(1 + x^{1+2d}\right)^3}\end{aligned} \tag{3.24}$$

for $\frac{\pi}{4} \le \varphi \le \frac{\pi}{2}$ and $x \ge 100$. Notice that $\frac{\partial}{\partial\varphi} f_3(x,\varphi) \ge 0$ for $0 \le \varphi \le \frac{\pi}{2}$ and $x \ge 100$. For f_4 we have

$$\begin{aligned}\frac{\partial}{\partial\varphi} f_4(x,\varphi) =& \frac{\left(x^{1+2d} - 1\right)\log^2(x)\left(\left(x^{\frac{3}{2}+3d} + x^{\frac{1}{2}+d}\right)\cos(\varphi) + 2x^{1+2d}\cos^2(\varphi) - 4x^{1+2d}\right)}{\left(1 - 2x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)^3} \\ \ge& -\frac{4x^{1+2d}\left(x^{1+2d} - 1\right)\log^2(x)}{\left(1 - x^{\frac{1}{2}+d}\right)^6}\end{aligned} \tag{3.25}$$

if $0 \le \varphi \le \frac{\pi}{2}$ and $x \ge 100$. Furthermore, notice that because $\frac{1}{\sqrt{2}} x^{\frac{1}{2}+d} - 3 \ge 0$ for $x \ge 100$ we may deduce from equation (3.25) that the derivative is positive for $0 \le \varphi \le \frac{\pi}{4}$ and $x \ge 100$. In the range $\frac{\pi}{4} \le \varphi \le \frac{\pi}{2}$ we have

$$\begin{aligned}\frac{\partial}{\partial\varphi} \frac{d^2 a_2}{2} f_3(x,\varphi) \ge& \frac{d^2 a_2}{2} \frac{\log^2(x)\left(x^{\frac{5}{2}+5d} - 6x^{\frac{3}{2}+3d} + x^{\frac{1}{2}+d}\right)}{\sqrt{2}\left(1 + x^{1+2d}\right)^3} \\ \ge& \frac{d^2 a_2}{4\sqrt{2}} \left(\frac{10000}{10001}\right)^3 \log^2(x) x^{-\left(\frac{1}{2}+d\right)} \\ \ge& 10 \log^2(x) x^{-(1+2d)}\end{aligned}$$

for $\varphi \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ and $x \ge 100$. Here we use $d \ge 0.7$ and $a_2 \ge 0.5$. Similarly, we have

$$\begin{aligned}\frac{\partial}{\partial\varphi} \frac{a_3}{2} f_4(x,\varphi) \ge& -\frac{a_3}{2} \frac{4x^{1+2d}\left(x^{1+2d} - 1\right)\log^2(x)}{\left(1 - x^{\frac{1}{2}+d}\right)^6} \\ \ge& -4\left(\frac{100}{99}\right)^6 \log^2(x) x^{-(1+2d)} \ge -5 \log^2(x) x^{-(1+2d)}\end{aligned}$$

for $\frac{\pi}{4} \le \varphi \le \frac{\pi}{2}$ and $x \ge 100$. Furthermore, we assumed $a_3 \le 2$. Therefore, we obtain that

$$\frac{\partial}{\partial\varphi}\left(\frac{d^2 a_2}{2} f_3(x,\varphi) + \frac{a_3}{2} f_4(x,\varphi)\right) \ge 0 \tag{3.26}$$

for $\frac{\pi}{4} \le \varphi \le \frac{\pi}{2}$ and $x \ge 100$. Altogether, we obtain from equations (3.19), (3.21), (3.22), (3.24), and (3.25) that $\frac{\partial}{\partial\varphi} q_1(x,\varphi) \ge 0$ for $0 \le \varphi \le \frac{\pi}{4}$ and $x \ge 100$. Equations (3.19), (3.23), and (3.26) imply that

$\frac{\partial}{\partial \varphi} q_1(x,\varphi) \geq 0$ for $\frac{\pi}{4} \leq \varphi \leq \frac{\pi}{2}$ and $x \geq 100$. Therefore, $q_1(x,\varphi)$ is maximized in the interval $\varphi \in \left[\frac{\pi}{2}, \pi\right]$ if $x \geq 100$. Now we may consider the derivatives of $f_i(x,\varphi)$ with respect to x and $\varphi \in \left[\frac{\pi}{2}, \pi\right]$. For f_1 we obtain

$$\begin{aligned}\frac{\partial}{\partial x} f_1(x,\varphi) =& \frac{x^{3d}\cos(\varphi) - (1+2d)\left(x^{\frac{1}{2}+5d} + x^{d-\frac{1}{2}}\right) + \left(\frac{1}{2}+2d\right)\left(x^{\frac{1}{2}+4d} + x^{2d-\frac{1}{2}}\right)}{\left(1 - 2x^{\frac{1}{2}+2d}\cos(\varphi) + x^{1+4d}\right)\left(1 - x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)} \sin(\varphi) \\ \leq& - \frac{(1+2d)\left(x^{\frac{1}{2}+5d} + x^{d-\frac{1}{2}}\right) - \left(\frac{1}{2}+2d\right)\left(x^{\frac{1}{2}+4d} + x^{2d-\frac{1}{2}}\right)}{\left(1 - 2x^{\frac{1}{2}+2d}\cos(\varphi) + x^{1+4d}\right)\left(1 - x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)} \sin(\varphi) \leq 0\end{aligned} \tag{3.27}$$

for $x \geq 100$, when $\varphi \in \left[\frac{\pi}{2}, \pi\right]$. Similarly, we find that

$$\begin{aligned}\frac{\partial}{\partial x} f_2(x,\varphi) =& \frac{1 - ((1+2d)\log(x) - 1)x^{1+2d}}{x\left(1 - 2x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)^2} \\ &+ \frac{\left(\left(\frac{1}{2}+d\right)\log(x) - 1\right)x^{\frac{3}{2}+3d} + 2x^{1+2d}\cos(\varphi) + \left(\left(\frac{1}{2}+d\right)\log(x) - 3\right)x^{\frac{1}{2}+d}}{x\left(1 - 2x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)^2}\cos(\varphi) \\ \leq& \frac{\left(\left(\frac{1}{2}+d\right)\log(x) - 1\right)x^{\frac{3}{2}+3d} - 2x^{1+2d} + \left(\left(\frac{1}{2}+d\right)\log(x) - 3\right)x^{\frac{1}{2}+d}}{x\left(1 - 2x^{\frac{1}{2}+d}\cos(\varphi) + x^{1+2d}\right)^2}\cos(\varphi) \leq 0\end{aligned} \tag{3.28}$$

for $\frac{\pi}{2} \leq \varphi \leq \pi$ and $x \geq 100$. For f_3 we obtain

$$\begin{aligned}\frac{\partial}{\partial x} f_3(x,\varphi) =& \frac{x^{d-\frac{1}{2}}\log^2(x)(1+2d)\left(\cos^2(\varphi) - 2\right)\left(x^{\frac{3}{2}+3d} - x^{\frac{1}{2}+d}\right)}{\left(1 - 2\cos(\varphi)x^{\frac{1}{2}+d} + x^{1+2d}\right)^3} \\ &+ \frac{4x^{d-\frac{1}{2}}\log(x)\left(\cos^2(\varphi) + 1\right)\left(x^{\frac{3}{2}+3d} + x^{\frac{1}{2}+d}\right)}{\left(1 - 2\cos(\varphi)x^{\frac{1}{2}+d} + x^{1+2d}\right)^3} \\ &+ \frac{x^{d-\frac{1}{2}}\log(x)\cos(\varphi)\left((1+2d)\left(x^{2+4d} - 1\right)\log(x) - 4x^{2+4d} - 24x^{1+2d} - 4\right)}{2\left(1 - 2\cos(\varphi)x^{\frac{1}{2}+d} + x^{1+2d}\right)^3} \\ \leq& \frac{x^{d-\frac{1}{2}}\log(x)\left(8\left(x^{\frac{3}{2}+3d} + x^{\frac{1}{2}+d}\right) - (1+2d)\log(x)\left(x^{\frac{3}{2}+2d} - x^{\frac{1}{2}+d}\right)\right)}{\left(1 - 2\cos(\varphi)x^{\frac{1}{2}+d} + x^{1+2d}\right)^3} \\ \leq& 0\end{aligned} \tag{3.29}$$

for $\frac{\pi}{2} \le \varphi \le \pi$ and $x \ge 100$. Lastly, we obtain

$$\begin{aligned}
\frac{\partial}{\partial x} f_4(x,\varphi) =& -\frac{\left(\frac{1}{2}+d\right)\sin(\varphi)\log^2(x)\left(2\cos(\varphi)\left(x^{\frac{3}{2}+3d}+x^{\frac{1}{2}+d}\right)+x^{2+4d}-6x^{1+2d}+1\right)}{x^{\frac{1}{2}-d}\left(1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d}\right)^3} \\
&-\frac{\sin(\varphi)\log(x)\left(4\cos(\varphi)\left(x^{\frac{3}{2}+3d}-x^{\frac{1}{2}+d}\right)-2\left(x^{2+4d}-1\right)\right)}{x^{\frac{1}{2}-d}\left(1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d}\right)^3} \\
\le& -\frac{\sin(\varphi)\log(x)\left(\left(\left(\frac{1}{2}+d\right)\log(x)-2\right)x^{2+4d}-((1+2d)\log(x)+4)x^{\frac{3}{2}+3d}\right)}{x^{\frac{1}{2}-d}\left(1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d}\right)^3} \\
&-\frac{\sin(\varphi)\log(x)\left(-6\left(\frac{1}{2}+d\right)\log(x)x^{1+2d}+((1+2d)\log(x)-4)x^{\frac{1}{2}+d}\right)}{x^{\frac{1}{2}-d}\left(1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d}\right)^3} \\
&-\frac{\sin(\varphi)\log(x)\left(\left(\frac{1}{2}+d\right)\log(x)+2\right)}{x^{\frac{1}{2}-d}\left(1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d}\right)^3} \\
\le& 0
\end{aligned} \tag{3.30}$$

for $\frac{\pi}{2} \le \varphi \le \pi$ and $x \ge 100$. Notice that because of equations (3.27),(3.28),(3.29), and (3.30) we know that the functions $f_i(x,\varphi)$ are decreasing in x when they are maximized in φ and $x \ge 100$. With the numerical verification of (3.18) from before, we have now established this inequality for all integer $\alpha \ge 2$ and $m \ge 1$. With equation (3.18) and the fact that over each prime can only lie n_K prime ideals in K [Neu99, proposition 8.2] we deduce that

$$E_u(\zeta_{\mathrm{K}})(T,d) \le n_K \sum_{p\in\mathbb{P}} \max_{\varphi} q_1(p,\varphi). \tag{3.31}$$

So we only need to evaluate the remaining prime series. To this end, we first find a majorant, which we evaluate by its connection to the Riemann zeta function. We will use this to evaluate the tail of the infinite series in (3.31). We evaluate the contribution of the remaining terms below the cut-off numerically. We again consider the individual terms of q_1 separately. For the first term, we obtain that

$$\begin{aligned}
&\frac{4}{\pi}\arctan\left(\frac{\sin(\varphi)}{x^{\frac{1}{2}+d}-\cos(\varphi)}\right)-\frac{2}{\pi}\arctan\left(\frac{\sin(\varphi)}{x^{\frac{1}{2}+2d}-\cos(\varphi)}\right) \\
\le&\frac{4}{\pi}\arctan\left(x^{-\frac{1}{2}-d}\right)-\frac{2}{\pi}\arctan\left(x^{-\frac{1}{2}-2d}\right) \\
\le&\frac{2}{\pi}\left(\log\left(1+x^{-\frac{1}{2}-d}\right)-\log\left(1-x^{-\frac{1}{2}-d}\right)\right)-\frac{1}{\pi}\left(\log\left(1+x^{-\frac{1}{2}-2d}\right)-\log\left(1-x^{-\frac{1}{2}-2d}\right)\right) \\
=:&u_1(x)
\end{aligned} \tag{3.32}$$

for $\frac{\pi}{2} \le \varphi \le \pi$ and $x \ge 100$. The first inequality follows from observing that the φ-derivative of f_1 given in (3.20) is negative for $\varphi \in \left[\frac{\pi}{2},\pi\right]$ and large x. The second inequality follows from the taylor series for the arctan(·) and the log(·) functions. For the second term, we find that

$$\frac{da_1}{2}\frac{\log(x)(1-x^{\frac{1}{2}+d}\cos(\varphi))}{1-2x^{\frac{1}{2}+d}\cos(\varphi)+x^{1+2d}} \leq \frac{da_1}{2}\frac{\log(x)}{1+x^{\frac{1}{2}+d}} =: u_2(x), \tag{3.33}$$

when $\frac{\pi}{2} \leq \varphi \leq \pi$ and $x \geq 100$. Lastly, we may estimate the remaining terms by

$$\begin{aligned}
\frac{d^2a_2}{2}f_3(x,\varphi)+&\frac{a_3}{2}f_4(x,\varphi) \\
&= \frac{d^2a_2}{2}\operatorname{Re}\left(\frac{\log^2(x)x^{\frac{1}{2}+d+\frac{i\varphi}{\log(x)}}}{\left(1-x^{\frac{1}{2}+d+\frac{i\varphi}{\log(x)}}\right)^2}\right) + \frac{a_3}{2}\operatorname{Im}\left(\frac{\log^2(x)x^{\frac{1}{2}+d+\frac{i\varphi}{\log(x)}}}{\left(1-x^{\frac{1}{2}+d+\frac{i\varphi}{\log(x)}}\right)^2}\right) \\
&\leq c\left|\frac{\log^2(x)x^{\frac{1}{2}+d+\frac{i\varphi}{\log(x)}}}{\left(1-x^{\frac{1}{2}+d+\frac{i\varphi}{\log(x)}}\right)^2}\right| = c\frac{\log^2(x)x^{\frac{1}{2}+d}}{1-2\cos(\varphi)x^{\frac{1}{2}+d}+x^{1+2d}} \leq c\frac{\log^2(x)x^{\frac{1}{2}+d}}{1+x^{1+2d}} \\
&\leq c\left(\frac{1}{2}\frac{\log^2(x)x^{\frac{1}{2}+d}}{(1-x^{\frac{1}{2}+d})^2}+\frac{1}{2}\frac{\log^2(x)x^{\frac{1}{2}+d}}{(1+x^{\frac{1}{2}+d})^2}\right) =: u_3(x),
\end{aligned} \tag{3.34}$$

where $c = c_{d,a_2,a_3} := \max_{s\in\mathbb{C}} \frac{\frac{d^2a_2}{2}\operatorname{Re}(s)+\frac{a_3}{2}\operatorname{Im}(s)}{|s|} = \frac{1}{2}\sqrt{(d^2a_2)^2+a_3^2}$, as well as $\frac{\pi}{2} \leq \varphi \leq \pi$ and $x \geq 100$. We use equations (3.17), (3.32), (3.33), and (3.34) to deduce that

$$\begin{aligned}
\sum_{p\in\mathbb{P},p>M}\max_{\varphi} q_1(p,\varphi) \leq &\sum_{p\in\mathbb{P},p>M} u_1(p)+u_2(p)+u_3(p) \\
=&\frac{2}{\pi}\left(2\log\zeta\left(\frac{1}{2}+d\right)-\log\zeta(1+2d)\right) \\
&-\frac{1}{\pi}\left(2\log\zeta\left(\frac{1}{2}+2d\right)-\log\zeta(1+4d)\right) \\
&+\frac{da_1}{2}\left(2\frac{\zeta'}{\zeta}(1+2d)-\frac{\zeta'}{\zeta}\left(\frac{1}{2}+d\right)\right) \\
&+c\left(\left(\frac{\zeta''}{\zeta}-\left(\frac{\zeta'}{\zeta}\right)^2\right)\left(\frac{1}{2}+d\right)-2\left(\frac{\zeta''}{\zeta}-\left(\frac{\zeta'}{\zeta}\right)^2\right)(1+2d)\right) \\
&-\sum_{p\in\mathbb{P},p\leq M} u_1(p)+u_2(p)+u_3(p) \\
\leq&1.86298,
\end{aligned}$$

where $M = 10^5$. Computing the remaining terms in equation (3.31) numerically yields

$$\sum_{p\in\mathbb{P},p\leq M}\max_{\varphi} q_1(p,\varphi) \leq 6.04758.$$

Altogether, we obtain that $E_u(\zeta_{\mathrm{K}})(T,d) \leq 7.911n_K$. We obtain the lower bound for $E_l(\zeta_{\mathrm{K}})(T,d)$ from the upper bound by observing that it is exactly the odd part of $E_u(\zeta_{\mathrm{K}})(T,d)$ minus the even part of

$E_u(\zeta_{\mathrm{K}})(T,d)$. We have that

$$\begin{aligned}
E_l(\zeta_{\mathrm{K}})(T) = \sum_{\mathfrak{p}\subset\mathcal{O}_K} \frac{4}{\pi}\arctan\left(\frac{\sin\varphi}{\alpha^{\frac{1}{2}+d}-\cos\varphi}\right) - \frac{2}{\pi}\arctan\left(\frac{\sin\varphi}{\alpha^{\frac{1}{2}+2d}-\cos\varphi}\right)& \\
- \frac{da_1}{2}\log(\alpha)\frac{1-\alpha^{\frac{1}{2}+d}\cos\varphi}{1-2\alpha^{\frac{1}{2}+d}\cos\varphi+\alpha^{1+2d}}& \\
+ \frac{d^2a_2}{2}(\log(\alpha))^2\frac{\alpha^{\frac{1}{2}+d}((1+\alpha^{1+2d})\cos\varphi-2\alpha^{\frac{1}{2}+d})}{(1-2\alpha^{\frac{1}{2}+d}\cos\varphi+\alpha^{1+2d})^2}& \\
- \frac{a_3}{2}(\log(\alpha))^2\frac{\alpha^{\frac{1}{2}+d}(1-\alpha^{1+2d})\sin\varphi}{(1-2\alpha^{\frac{1}{2}+d}\cos(\varphi)+\alpha^{1+2d})^2}& \\
= \sum_{\mathfrak{p}\subset\mathcal{O}_K} q_2(\alpha,\varphi) \le \sum_{\mathfrak{p}\subset\mathcal{O}_K} \max_{\varphi} q_2(\alpha,\varphi),&
\end{aligned}$$

where $\alpha := N\mathfrak{p}$ and $\varphi := T\log(N\mathfrak{p})$. Then

$$\min_{\varphi} q_2(\alpha,\varphi) = -\max_{\varphi}(-q_2(\alpha,\varphi)) = q_2(\alpha,\varphi_0) = -q_1(\alpha,-\varphi_0) \ge -\max_{\varphi} q_1(\alpha,\varphi),$$

where φ_0 is a maximizer of $-q_2(\alpha,\varphi)$. Hence, the minimum of $q_2(\alpha,\varphi)$ is lower bounded by minus the maximum of $q_1(\alpha,\varphi)$ and we may lower bound $E_l(\zeta_{\mathrm{K}})(T)$ by the negative of the upper bound for $E_u(\zeta_{\mathrm{K}})(T)$. This concludes the proof of the lemma. □

Combining Lemmas 3.4, 3.5, 3.7 and 3.8 yields the following inequalities. We have

$$\begin{aligned}
E_u(\xi_K)(T) \le& \frac{T}{\pi}\log\left(d_K\left(\frac{T}{2\pi e}\right)^{n_K}\right) + \frac{da_1}{4}\log\left(d_K\left(\frac{T}{2\pi}\right)^{n_K}\right) + 8.003n_K + 2.001, \\
E_l(\xi_K)(T) \ge& \frac{T}{\pi}\log\left(d_K\left(\frac{T}{2\pi e}\right)^{n_K}\right) - \frac{da_1}{4}\log\left(d_K\left(\frac{T}{2\pi}\right)^{n_K}\right) - 8.161n_K + 1.547
\end{aligned}$$

for $T \ge 1$. Plugging in $d = 0.722$ and $a_1 = 1.07$ then yields

$$\begin{aligned}
\left|N_K(T) - \frac{T}{\pi}\log\left(d_K\left(\frac{T}{2\pi e}\right)^{n_K}\right)\right| \le& 0.194\log\left(d_K\left(\frac{T}{2\pi}\right)^{n_K}\right) + 8.161n_K + 2.001 \\
=& 0.194(\log d_K + n_K\log T) + 8.161n_K + 2.001
\end{aligned}$$

for $T \ge 1$, where T satisfies $\zeta_{\mathrm{K}}(x+iT) \ne 0$ for all $x \in \mathbb{R}$. Theorem 2.1 follows from this and equation (3.1).

Remark 3.9. Table (4) contains different choices for C_1, C_2, C_3 such that

$$\left|N_K(T) - \frac{T}{\pi}\log\left(d_K\left(\frac{T}{2\pi e}\right)^{n_K}\right)\right| \le C_1\log(T) + C_2 n_K + C_3 \tag{3.35}$$

for the different tuples (d, a_1, a_2, a_3) from (3).

i	C_1	C_2	C_3
1	0.2	6.803	2.001
2	0.24	4.155	2.001
3	0.28	3.055	2.001
4	0.32	2.447	2.001
5	0.36	2.061	2.001
6	0.4	1.792	2.001

Table 4: Further admissible choices for (C_1, C_2, C_3) in equation (3.35)

We find that these improve on the results in table (1)[HSW21] in every constant. Table (5) compares the bounds from this method to the bounds in [HSW21] at the example of the Dedekind zeta-function associated with the field $K = \mathbb{Q}(i)$ with $n_K = 2$ and absolute discriminant $d_K = 4$.

T	table (1) [HSW21]	table (4)
10	12.285	7.982
10^3	15.210	11.596
10^6	19.298	16.181
10^9	23.386	20.105
10^{12}	27.474	23.907
10^{15}	31.488	27.223

Table 5: Comparison of the results from this paper with [HSW21]

Proof of Corollary 1.2. We apply Theorem 2.1 in the case where $K = \mathbb{Q}$, yielding that

$$\left| N_{\mathbb{Q}}(T) - \frac{T}{\pi} \log\left(\frac{T}{2\pi e}\right) \right| \leq 0.194 \log T + 10.162.$$

Now, we observe that due to the complex symmetry of the zeros of $\zeta(s)$ and because $\zeta(\sigma) \neq 0$ for all real $\sigma \in [0,1]$, $N_{\mathbb{Q}}(T) = 2N(T)$ for all $T \geq 0$. Hence,

$$\left| N(T) - \frac{T}{2\pi} \log\left(\frac{T}{2\pi e}\right) \right| \leq 0.097 \log T + 5.081 \tag{3.36}$$

for $T \geq 1$, as desired. □

Remark 3.10. The following table contains different choices for C_1, C_2 such that

$$\left| N(T) - \frac{T}{2\pi} \log\left(\frac{T}{2\pi e}\right) \right| \leq C_1 \log(T) + C_2 \tag{3.37}$$

for the different tuples (d, a_1, a_2, a_3) from (3).

i	C_1	C_2	optimal range
1	0.1	4.402	$\left(e^{66.2}, e^{226,\overline{3}}\right)$
2	0.12	3.078	$\left(e^{27.5}, e^{66.2}\right)$
3	0.14	2.528	$\left(e^{15.2}, e^{27.5}\right)$
4	0.16	2.224	$\left(e^{9.65}, e^{15.2}\right)$
5	0.18	2.031	$\left(e^{6.75}, e^{9.65}\right)$
6	0.2	1.896	$\left(1, e^{6.75}\right)$

Table 6: Admissible choices for (C_1, C_2) for equation (3.37) with the corresponding ranges where each estimate is best.

Beyond $e^{226,\overline{3}}$, the bound in equation (3.36) gives the best result from this method. These bounds improve on the bounds in [BW25] for $T \geq 4.1$.

Remark 3.11. Notice that this method to estimate the error in the zero counting function of a L-function predominantly depends on Lemma 3.1. However, to apply this lemma, we simply require that $\frac{1}{2} + iT - \rho$ is within the critical strip. Hence, the method is principle amenable to any L-function in the Selberg class.

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